Core Cloud and Networking Skills

Day 1

QA

Learn. To Change.

Overview

Day 1
• Approaches to problem solving • Why use a task tracking system? • Maintenance approaches • Security concepts • Malware
Day 2
• Encryption • What is the cloud? • Virtualisation • Backup and storage
Day 3
• Wireless networking • Internet Protocol version 4 (IPv4)

Overview

Day 4
• Internet Protocol version 6 (IPv6) • Network services • Command Prompt
Day 5
• Troubleshooting • Critical analysis • Review and reflection

How to complete this course

This module covers practicals using Miro, GoToMyPC/LOD, and Cisco Packet Tracer.

Please ensure you have these installed and working on your own machine.

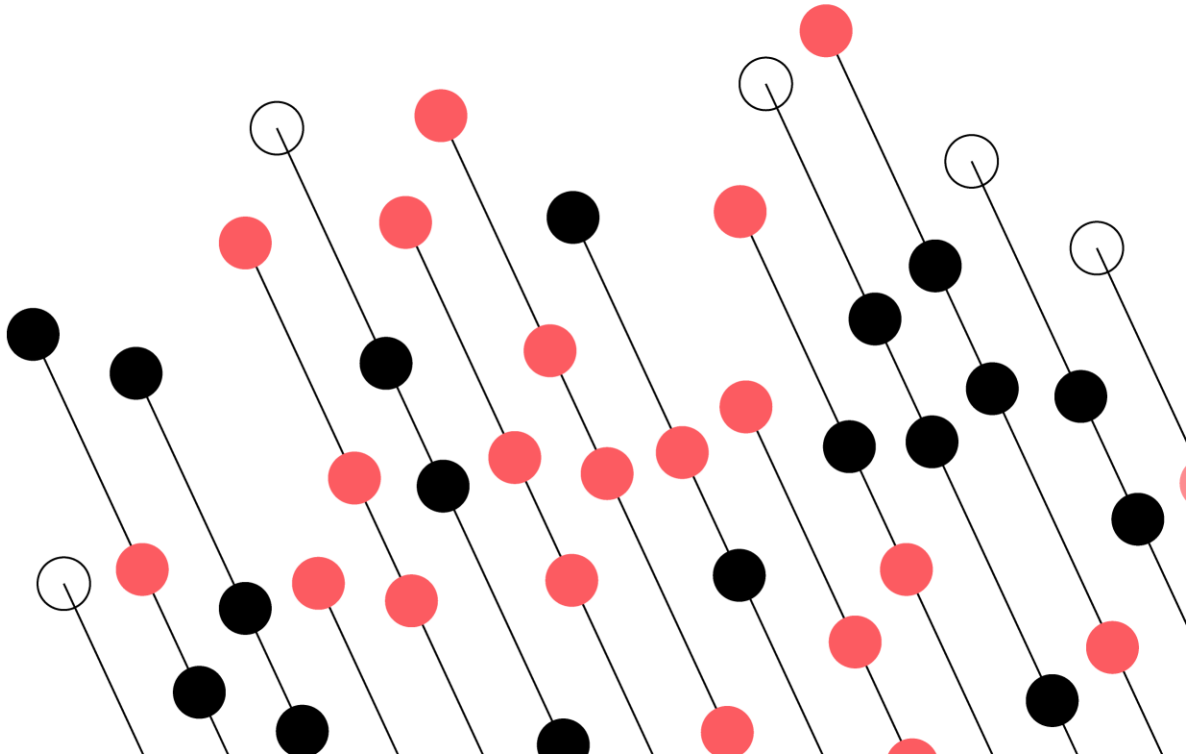
If you have any issues, please inform your tutor.

How to complete this module

As part of your Live event, you will have the sides and activity book to complete.

- Open your '**Exercise guide**' for this course and use it to document your answers.
- Add your **full name and date** on the **first page** of your **Exercise Guide**.
- Use this PowerPoint as a guide to take you through the course.
- You may need to ask/do some research to answer some of the criteria.
- Once your **Exercise Guide** is completed, upload it to **BUD** along with your **Certificate of completion**.
- And most importantly, if you need **help message or email your tutor**.

Approaches to problem solving



Approaches to problem solving

Problem-solving in the IT department involves identifying, analysing, and addressing challenges or issues with technology infrastructure, systems, and processes.

It’s essential skill for maintaining the efficient functioning of IT services, supporting system development, and ensuring effective maintenance.

Problem solving is a valuable skill that can enhance efficiency and productivity across various scenarios – not just the IT service desk.

Approaches to problem solving

This process follows a structured, eight-step approach:

1. **Identification:** Detecting the existence of a problem, such as a system outage, software bug, security vulnerability, or process inefficiency.

2. **Analysis:** Investigating the root cause by collecting data, replicating the issue, and assessing its impact on users and the organisation.

Approaches to problem solving

This process follows a structured, eight-step approach:

3. **Prioritisation:** Evaluating the problem's severity and urgency to determine the order in which it should be addressed.

4. **Solution design:** Creating a strategy to resolve the issue, which may involve coding fixes, system reconfiguration, new tool implementation, or process adjustments.

Approaches to problem solving

This process follows a structured, eight-step approach:

5. **Implementation:** Executing the solution, such as deploying patches, updating software, modifying hardware, or making procedural changes.

6. **Testing and validation:** Verifying that the solution effectively resolves the issue without causing additional problems.

Approaches to problem solving

This process follows a structured, eight-step approach:

7. **Documentation and review:** Logging the issue, the resolution process, and key takeaways to enhance future problem-solving efforts and prevent recurrence.

8. **Communication:** Keeping stakeholders informed about the issue's status, progress, and resolution.

Approaches to problem solving

There are various resources available to assist with problem solving:

- User guides
- Flowcharts
- Technical manuals
- Online forums
- Guidance from colleagues or subject matter experts

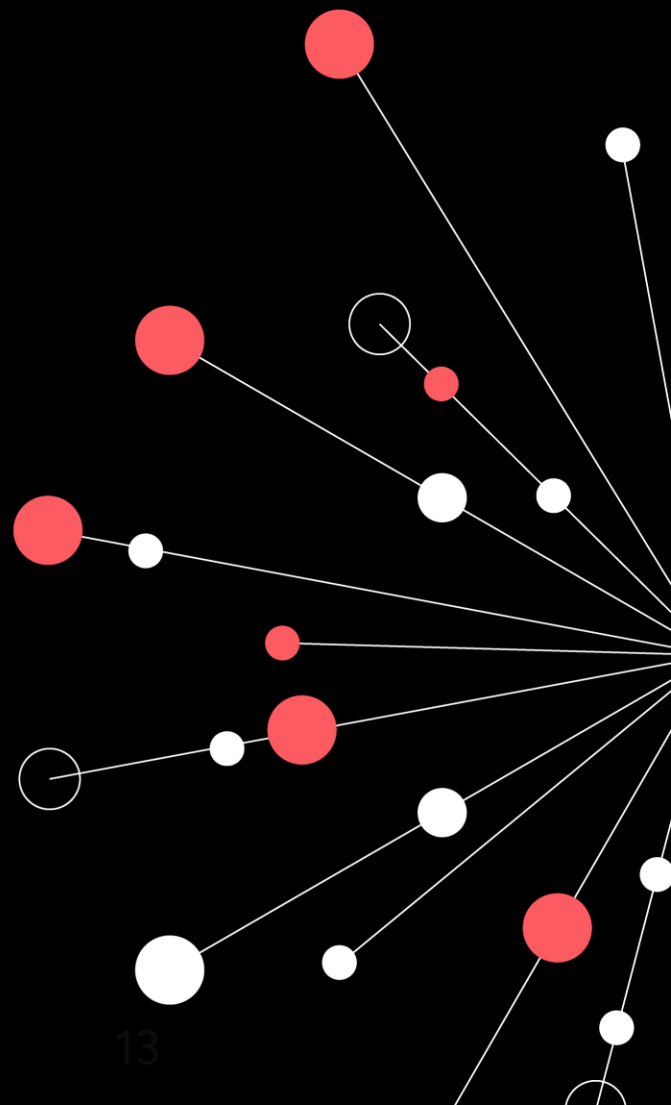
Each resource has its advantages and limitations. Carefully consider which one to use based on the situation.

For example, asking a colleague for help may also have drawbacks if it interrupts their workflow or is relied upon too frequently.

Activity 1: Research into problem solving

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



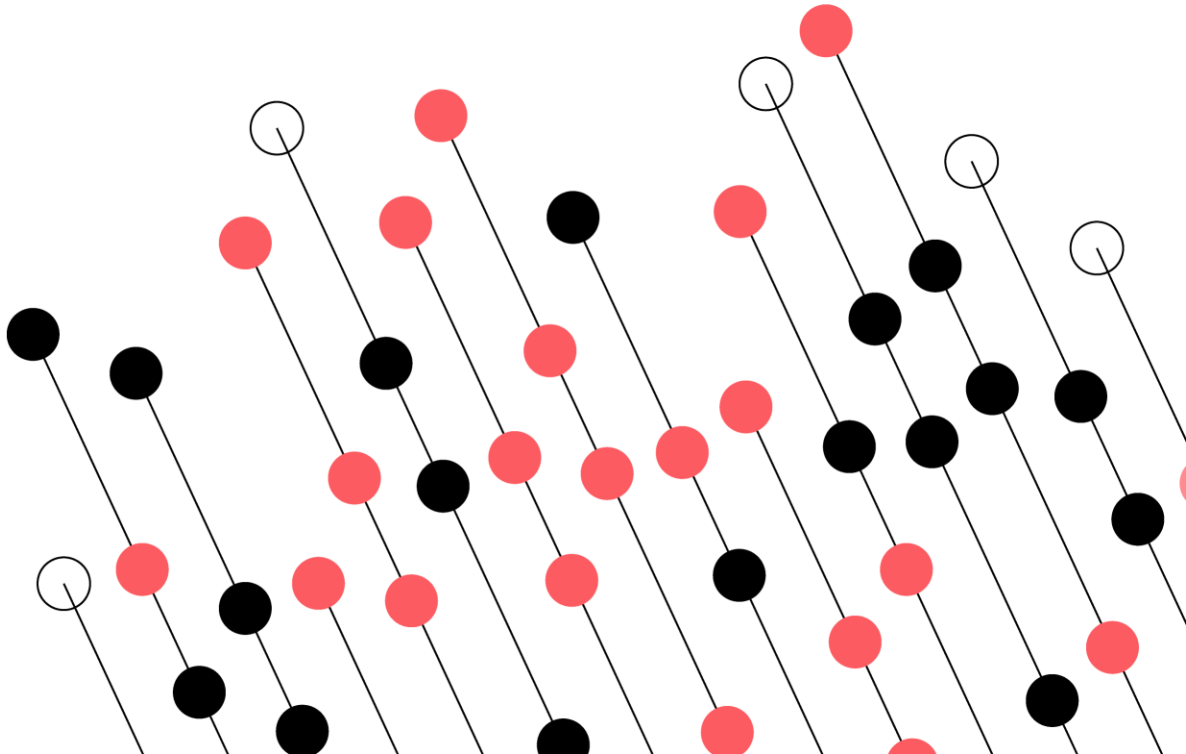
In summary: Approaches to problem solving

Problem-solving is a fundamental skill that involves identifying challenges, analysing potential causes, and implementing effective solutions to overcome them.

It requires critical thinking, creativity, and adaptability to navigate complex situations and uncertainties.

Effective problem-solving often follows a structured approach: defining the problem clearly, gathering relevant information, brainstorming solutions, evaluating options, and executing a plan.

Task tracking systems



Task tracking system

A task tracking system is a software tool or platform used to organise, oversee, and manage tasks, projects, and workflows associated with IT operations and services.

It ensures that all activities, whether simple or complex, are effectively assigned, monitored, and completed within designated timeframes.

Task tracking system

Key features of a task tracking system include the following:

- **Task management:** Allows for the creation, assignment, and categorisation of tasks based on priority, deadlines, and ownership.

- **Progress monitoring:** Tracks the status of tasks, providing real-time updates on whether a task is pending, in progress, or completed.

Task tracking system

Key features of a task tracking system include the following:

- **Collaboration tools:** Facilitates communication between team members through comments, file attachments, and shared updates.
- **Automation:** Automates repetitive tasks, reminders, and notifications to ensure efficiency and reduce manual follow-ups.

Task tracking system

Key features of a task tracking system include the following:

- **Reporting and analytics:** Generates insights into task performance, team productivity, and bottlenecks through detailed reports and dashboards.

- **Integration:** Seamlessly integrates with other IT tools, such as ticketing systems, communication platforms, and project management software.

Task tracking system

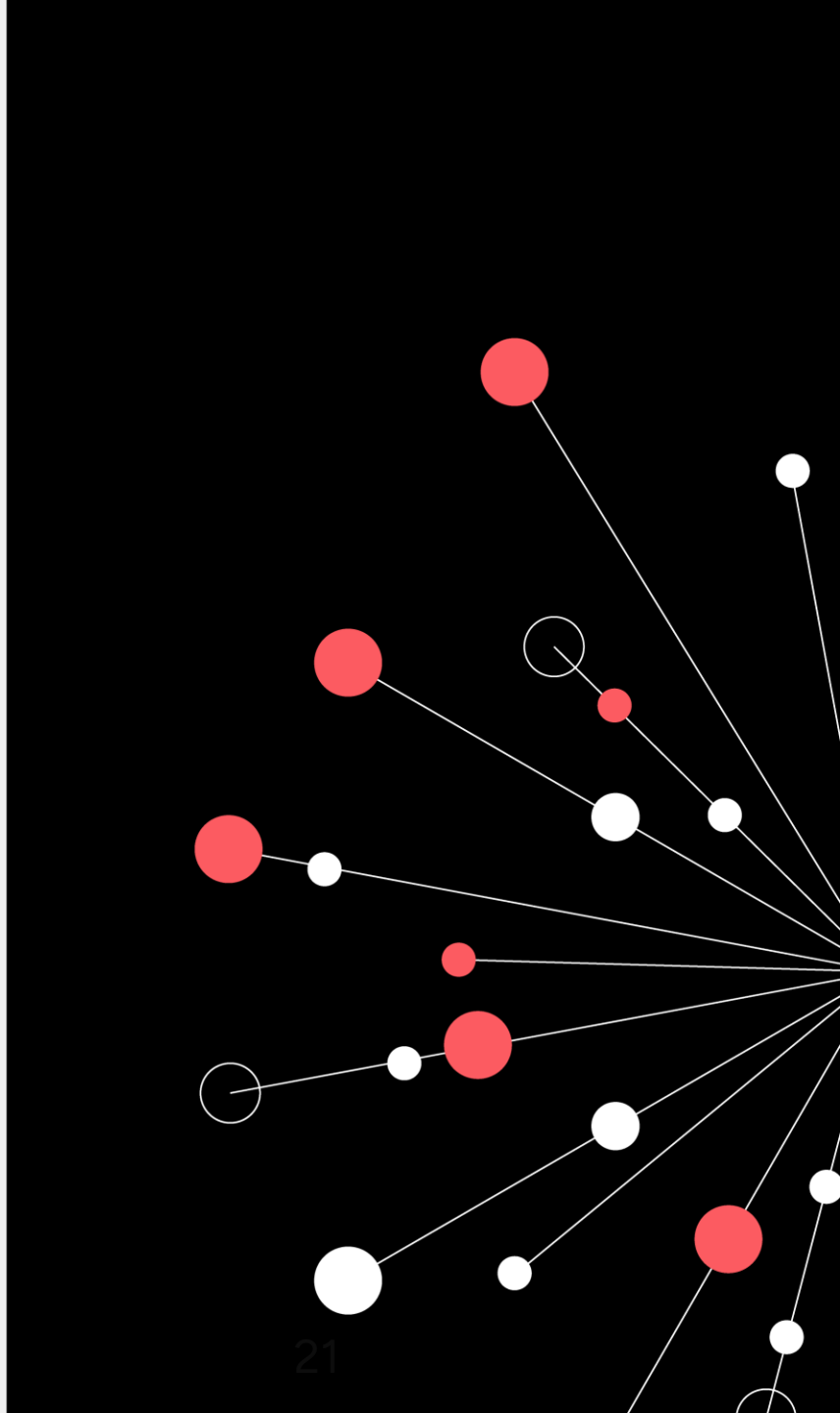
Some popular task tracking systems used in IT include the following:

- **Jira:** Commonly used for software development and IT service management.
- **Trello:** A visual task board ideal for team collaboration and tracking.
- **Asana:** Focused on project and task management across teams.
- **Monday.com:** Provides customisable workflows for task and project tracking.

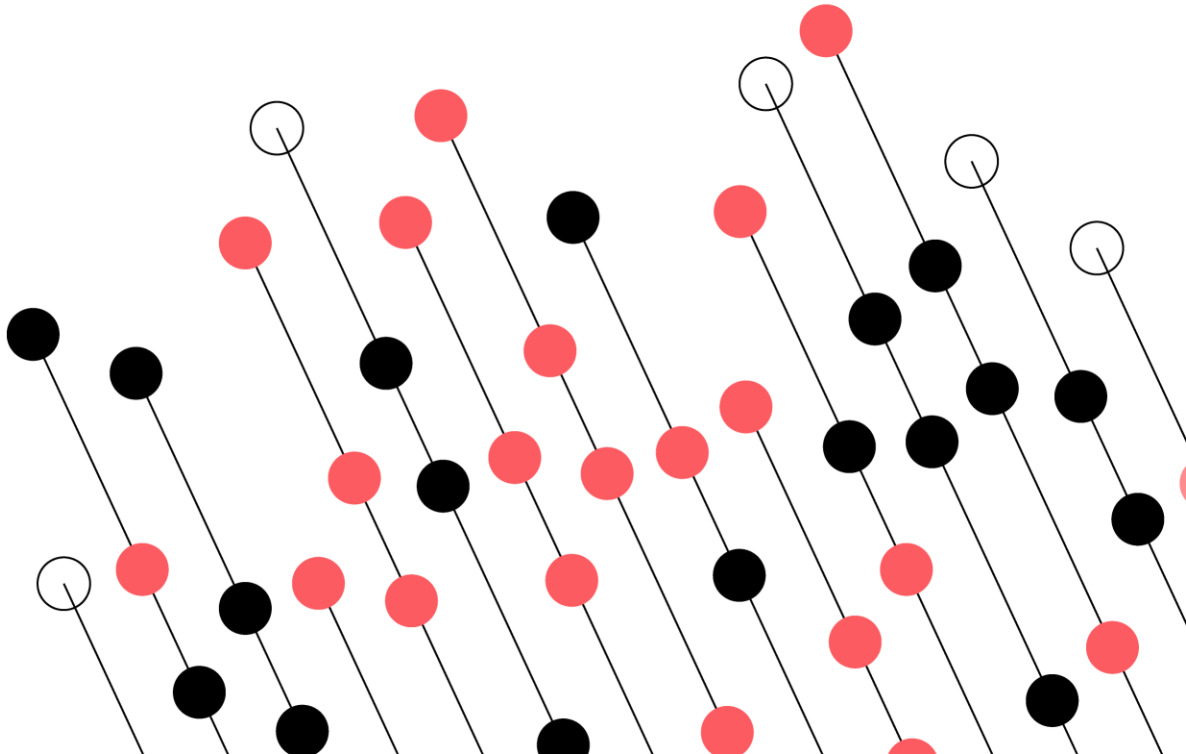
Activity 2: Task tracking systems

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Maintenance approaches



Maintenance approaches

Maintenance approaches are the strategies, techniques, and practices used to maintain the functionality, reliability, and durability of systems, equipment, or processes.

In IT and other technical fields, these approaches aim to resolve existing issues, address potential problems, and prevent future failures.

Maintenance approaches

Examples of IT maintenance include:

- fixing a server after it crashes or addressing a network outage after it has already occurred.
- using performance monitoring tools to track CPU usage or memory consumption and identifying when a server is likely to become overloaded.
- upgrading legacy systems or migrating to the cloud to avoid risks associated with aging hardware and unsupported software.
- responding to a cybersecurity breach or DDoS attack by shutting down affected services and implementing immediate containment measures.

Activity 3: Maintenance approaches

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.

Maintenance approaches

Preventive maintenance involves performing regular, scheduled inspections and maintenance activities to reduce the likelihood of equipment failure.

The goal is to ensure machinery and systems remain in optimal working condition and to address potential issues before they become significant problems.

Maintenance approaches

Predictive maintenance uses data, sensors, and advanced analytics to monitor the real-time condition of equipment.

Maintenance is performed only when specific indicators show that a component is likely to fail soon, allowing organisations to address issues proactively.

Maintenance approaches

Reactive maintenance (also called ‘run-to-failure’ maintenance) involves addressing equipment failures or breakdowns only after they occur.

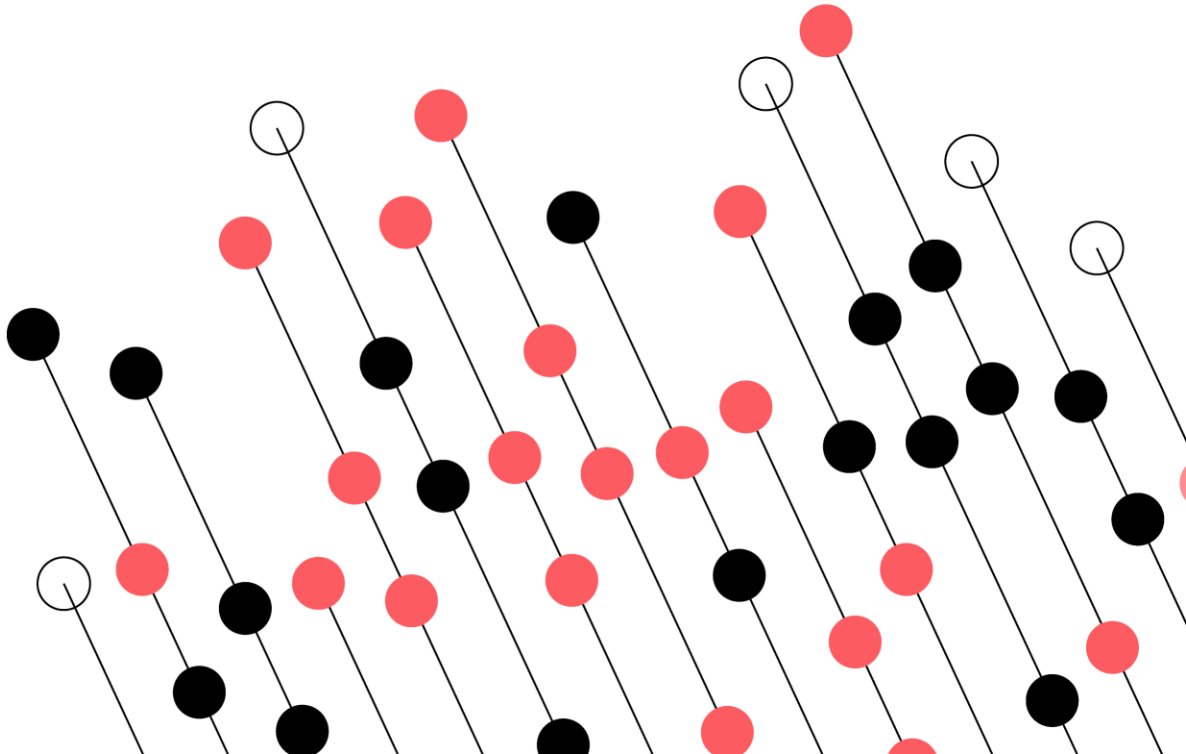
While it can minimise upfront costs, it often leads to higher costs in downtime and repairs.

In summary: Maintenance approaches

Effective maintenance approaches are essential for ensuring the reliability, safety, and efficiency of systems, equipment, and infrastructure.

By selecting and combining the right strategies based on operational needs and resources, organisations can optimise performance, reduce downtime, and extend the lifespan of their assets, ultimately contributing to cost savings and enhanced productivity.

Security concepts



Security concepts

Security in IT refers to the **practices, policies, and technologies** used to protect an organisation's **digital assets, systems, networks, and data** from **unauthorised access, attacks, damage, or theft.**

In today’s digital age, IT security is critical for safeguarding sensitive information, ensuring business continuity, and maintaining the integrity of systems.

Security concepts: CIA Triad

The CIA Triad is a framework in information security that outlines the essential principles for protecting data and systems:

1. Confidentiality
2. Integrity
3. Availability

It emphasises the protection of sensitive data, the accuracy of that data, and the continuous availability of systems and services.

Activity 4: CIA Triad

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

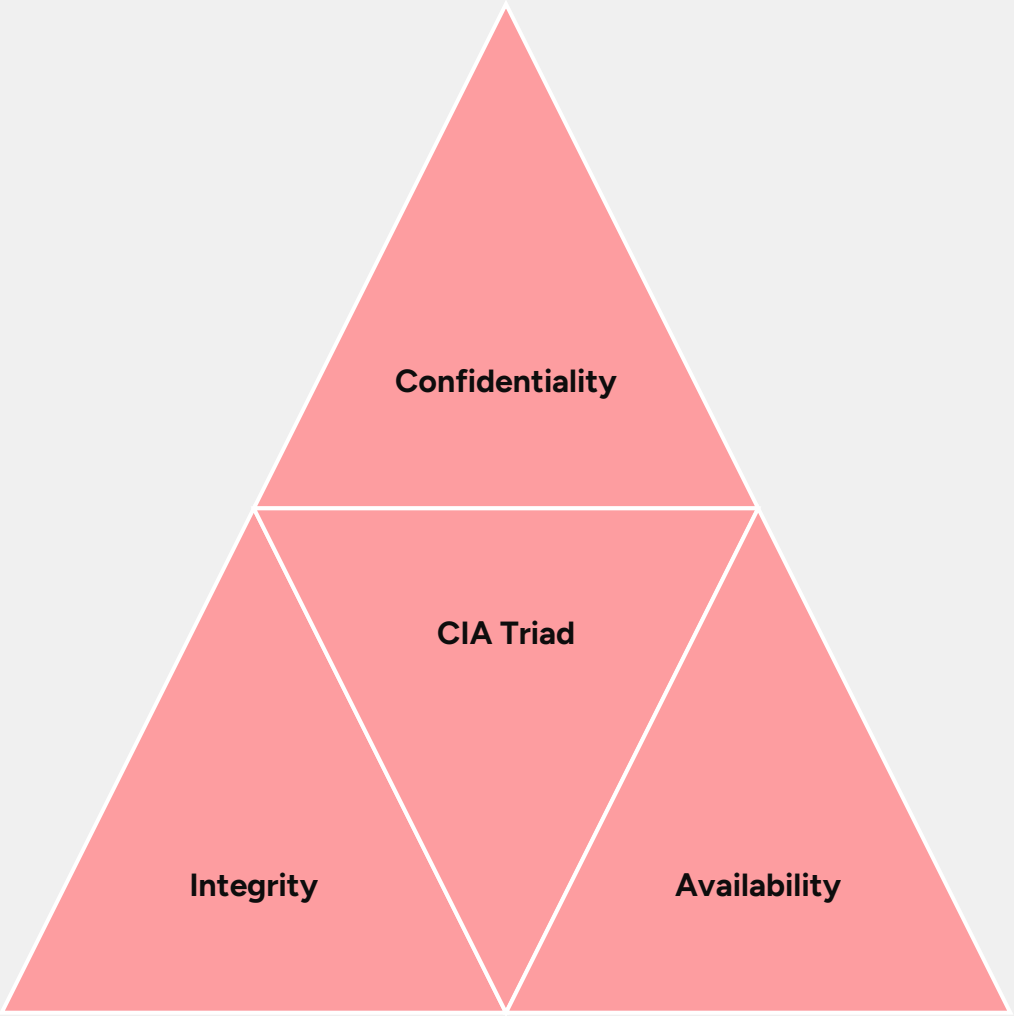
If you need any assistance just message the tutor.

Security concepts: CIA Triad

Confidentiality:
Ensure information isn't made available or disclosed to unauthorised individuals, entities, or processes.

Integrity
Protect the accuracy and completeness of assets and data.

Availability
Make sure that authorised users have access to the resources they need when they need them.



Security concepts: IAAA

IAAA is a fundamental set of principles that ensures only authorised individuals or systems can access resources, under appropriate conditions, while keeping a detailed record of all activities.

It stands for the following:

- Identity
- Authentication
- Authorisation
- Accounting

Security concepts: IAAA

- **Identification:** Uniquely identifies a user
- **Authentication:** Proof of identity
- **Authorisation:** Checking permissions for access to assets
- **Accountability:** Proof of who performed an action (also referred to as auditing)

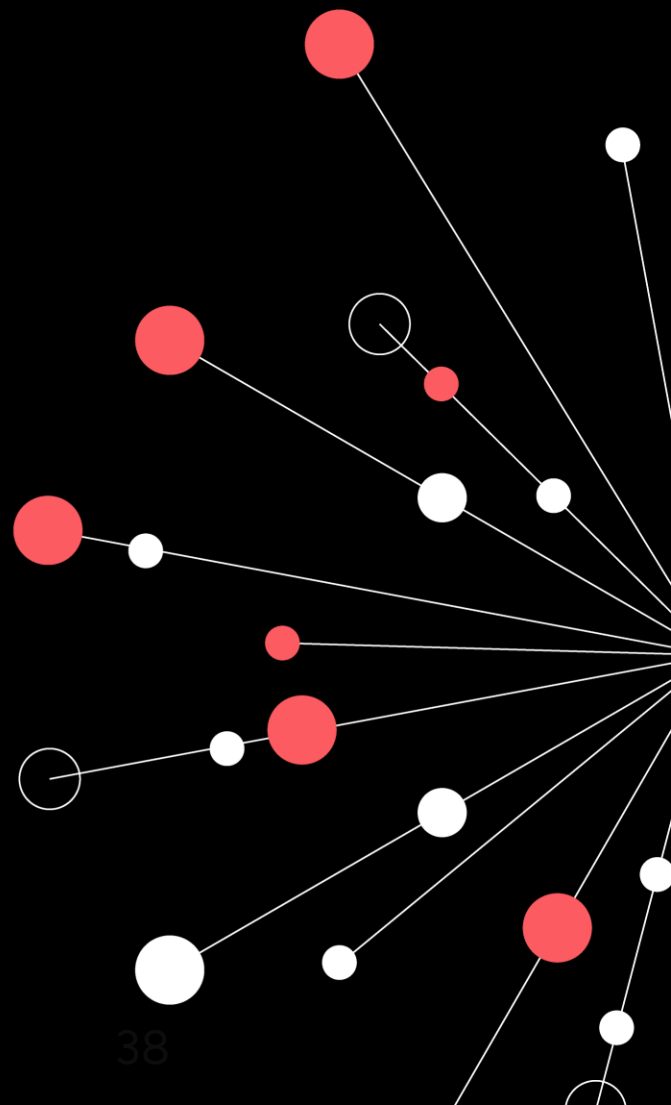
Security concepts: Comparing CIA Triad and IAAA

Aspect	CIA Triad	IAAA
Focus	Protecting data and systems	Managing identity and access control
Scope	Data security goals	Access and user / system behaviour management
Components	Confidentiality, integrity, availability	Identity, Authentication, Authorisation, Accounting
Example use case	Ensuring encrypted, verified, and available data	Implementing multi-factor authentication and logging activities

Activity 5: IAAA

Complete the activity within your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Security concepts: Identification

Associates a subject (user or software process) with an action performed on a network system (non-repudiation):

- Identifier
- Credentials
- Profile
- Issuance / enrolment
- Identity management

Security concepts: Authentication

Proves that a user or process is who it claims to be.

Five factors of authentication:

1. Something you know
2. Something you have
3. Something you are
4. Somewhere you are
5. Something you do

Security concepts: Authentication

Multi-factor authentication requires more than one type biometric technology:

- Fingerprint recognition
- Retinal scan / iris recognition
- Facial recognition

Security concepts: Authorisation

Granting users rights to resources:

- Subjects
- Objects
- Policy enforcement
- Ensuring only authorised rights are exercised
- Policy definition
- Determining rights

Security concepts: Accounting

Also known as auditing / logging:

- Provides account for actions
- Helps detect intrusions
- Appropriate selection of what to log
- Surveillance
- Incident reporting

Security concepts: Other concepts

- **Principle of least privilege:** Assign only permissions and rights needed to do the job but no more.

- **Defence in-depth strategy**: Multiple layers of defence to ensure breaches are difficult.

- **Explicit deny**: If a rule allowing an action does not exist, deny the action.

In summary: Security concepts

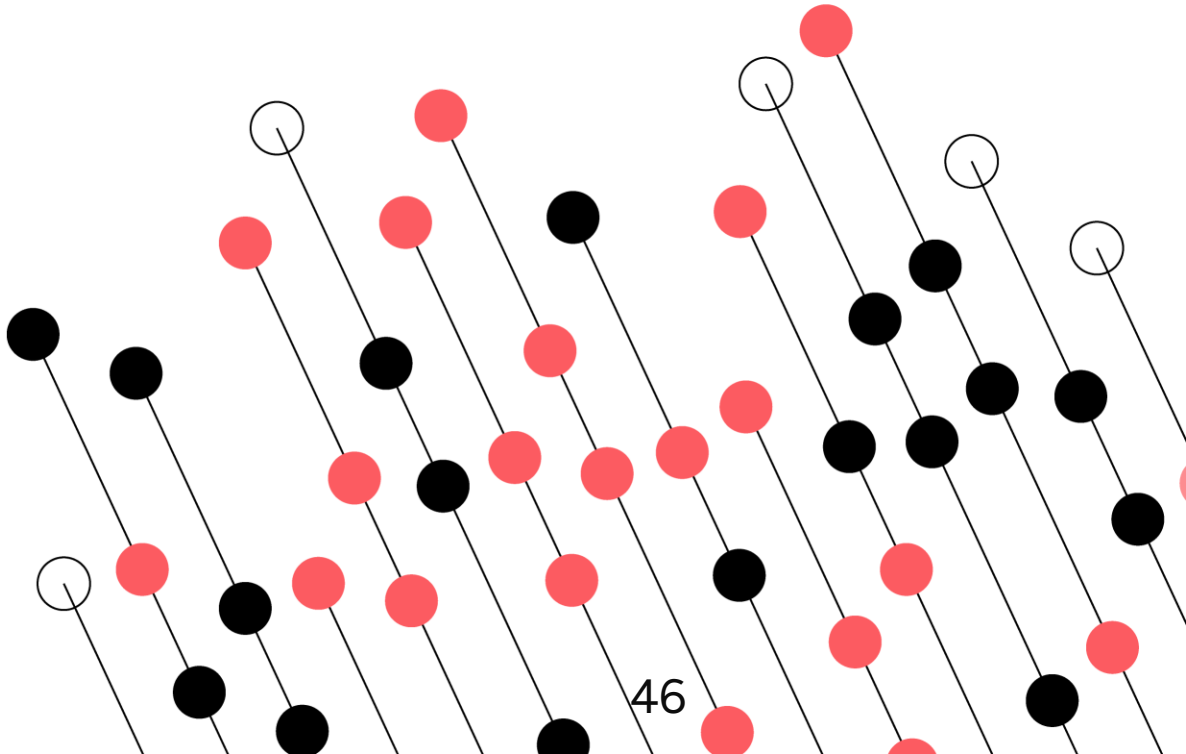
In conclusion, the CIA Triad and IAAA framework are fundamental security principles that cover different yet complementary facets of information protection.

The CIA Triad centres on the primary goals of safeguarding sensitive data from unauthorised access, alteration, and disruptions.

In contrast, the IAAA framework focuses on the processes and mechanisms that ensure system and data access is granted only to authenticated users, actions are governed by specific permissions, and all activities are properly logged for accountability.

Together, these concepts offer a comprehensive strategy for securing information, helping organisations create robust and resilient systems.

Malware



Malware

Malware is any type of software that is deliberately created to interfere with, harm, or gain unauthorised access to computer systems, networks, or sensitive data.

It covers a wide range of cyber threats, including various harmful programs that are designed with malicious intent.

Understanding malware is crucial for maintaining cybersecurity and protecting your digital assets.

Malware: Key points

Disruption: Malware can disrupt the normal functioning of a computer or network, causing it to slow down or crash.

Damage: It can damage files or systems, leading to data loss or corruption.

Unauthorised access: Malware often seeks to gain unauthorised access to sensitive information, such as personal data or financial details.

Variety of types: The category of malware includes viruses, worms, trojans, ransomware, and spyware, each with its own method of attack and impact.

Malware: Common symptoms and indicators

Sluggish performance: Malware consumes system resources, causing slowdowns or delays.

System crashes or freezes: Malware destabilises the system, leading to crashes or unresponsiveness.

Unusual pop-ups or ads: Unexpected ads or redirects may signal adware or malware.

Unauthorised changes: Malware may alter, delete, or rename files, or change system settings.

Increased activity: Unexplained disk or network activity suggests malware replication or communication.

Malware: Types

- Viruses
- Worms
- Trojans
- Ransomware
- Spyware
- Adware
- Rootkits
- Bots/botnets
- Keyloggers
- Logic bombs
- Backdoors
- Malware
- Scareware
- Exploit kits

Activity 6: Defining malware

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

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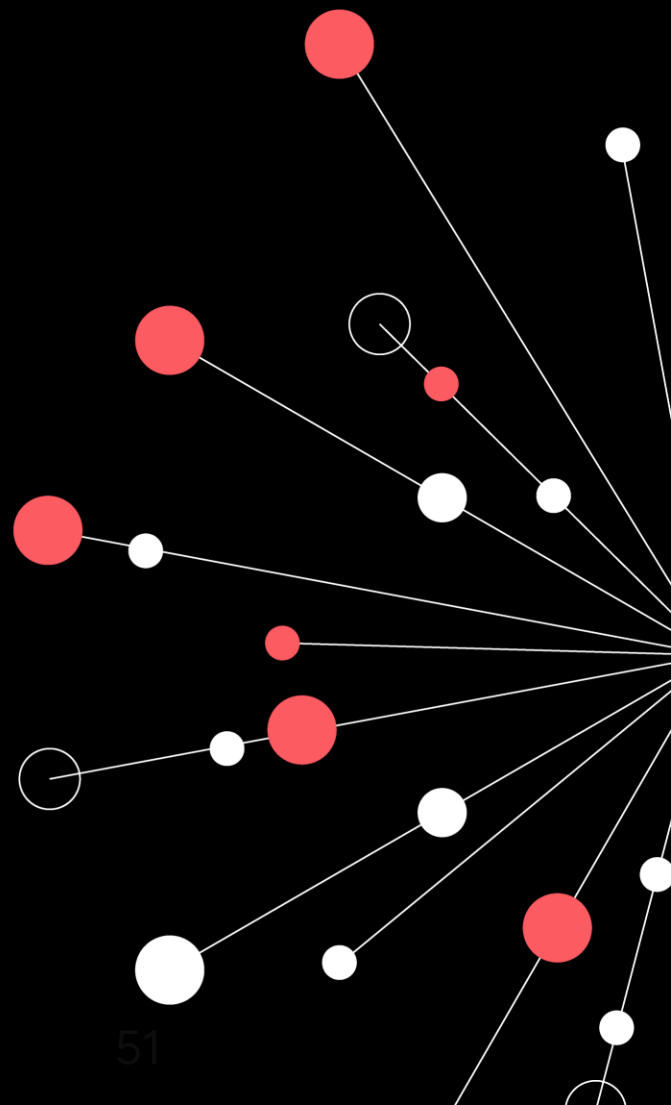

Malware prevention

Malware prevention involves implementing a comprehensive set of security measures designed to safeguard systems, networks, and data from malicious software.

This includes proactive steps to identify and block potential threats before they can cause harm, as well as reactive measures to mitigate damage in the event of an infection.

Malware prevention

Methods can include:

- installing and updating security software.
- enabling firewalls.
- keeping software and systems updated.
- using strong passwords and authentication.
- being cautious with emails and links.
- practicing safe browsing.
- backing up data regularly.
- limiting user privileges.

Malware prevention: Antivirus software

Antivirus software is a security program designed to detect, prevent, and remove malware from computers, networks, and devices.

It provides a critical line of defence against cyber threats to keep systems and data secure.

Malware prevention: Antivirus software

How it works:

- **Signature-based detection**: Compares files against a database of known malware signatures.
- **Heuristic analysis**: Detects suspicious or abnormal behaviour, even if no signature exists.
- **Sandboxing**: Runs suspicious files in a virtual environment to test for malicious behaviour without harming the actual system.

Malware prevention: Removing malware

Here are key methods to effectively remove malware:

- Use antivirus or anti-malware software.
- Use safe mode to run your system with minimal programs, preventing malware from running.
- Immediately disconnect from the internet to prevent malware from spreading or communicating with its source.
- Deleting temporary files to help remove malware or speed up scans.
- Some malware (e.g., ransomware or rootkits) requires specialised tools for removal.

Malware prevention: Firewalls

A firewall is a security system that monitors, filters, and controls incoming and outgoing network traffic based on predefined security rules.

It acts as a barrier between trusted internal networks and untrusted external networks, such as the internet, to prevent unauthorised access, cyberattacks, and the spread of malware.

Various types of firewalls include:

- packet filtering.
- stateful inspection.
- stateless filtering.

Activity 7: Malware

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

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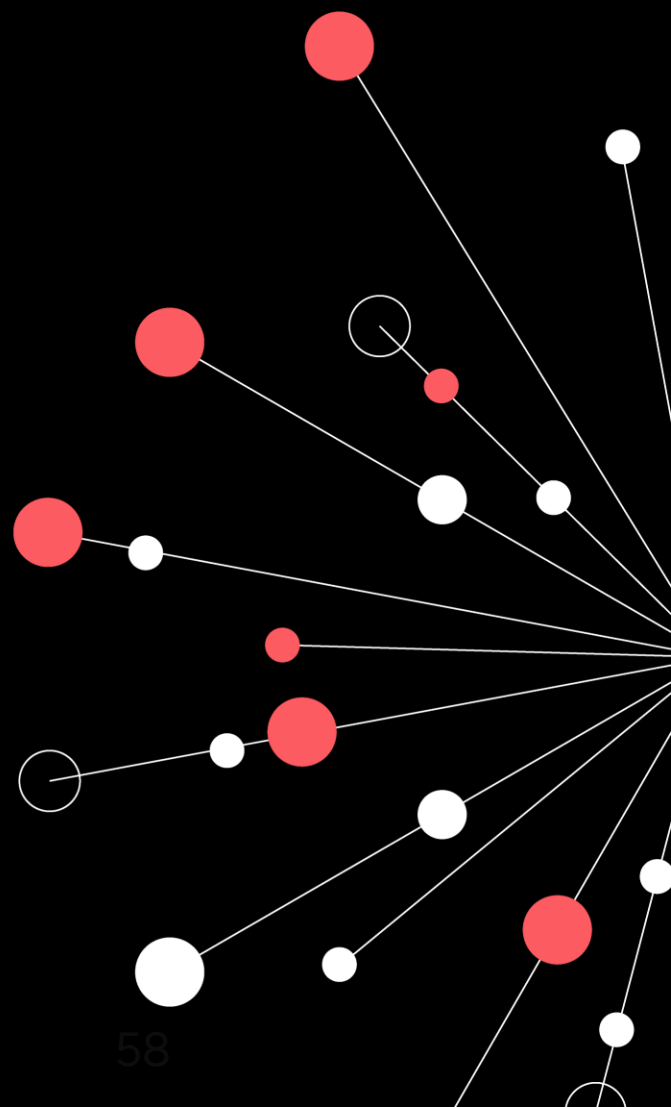


In summary: Malware

In summary, malware and antivirus software represent two sides of the cybersecurity landscape.

Malware, a collective term for malicious software such as viruses, worms, ransomware, and spyware, poses significant threats to individuals and organisations by compromising data, systems, and privacy.

Antivirus software serves as a critical defence mechanism, designed to detect, block, and remove malware. It employs techniques like signature-based detection, heuristic analysis, and real-time scanning to safeguard systems from known and emerging threats.

Choosing the right antivirus software involves balancing factors like performance, features, usability, and cost.

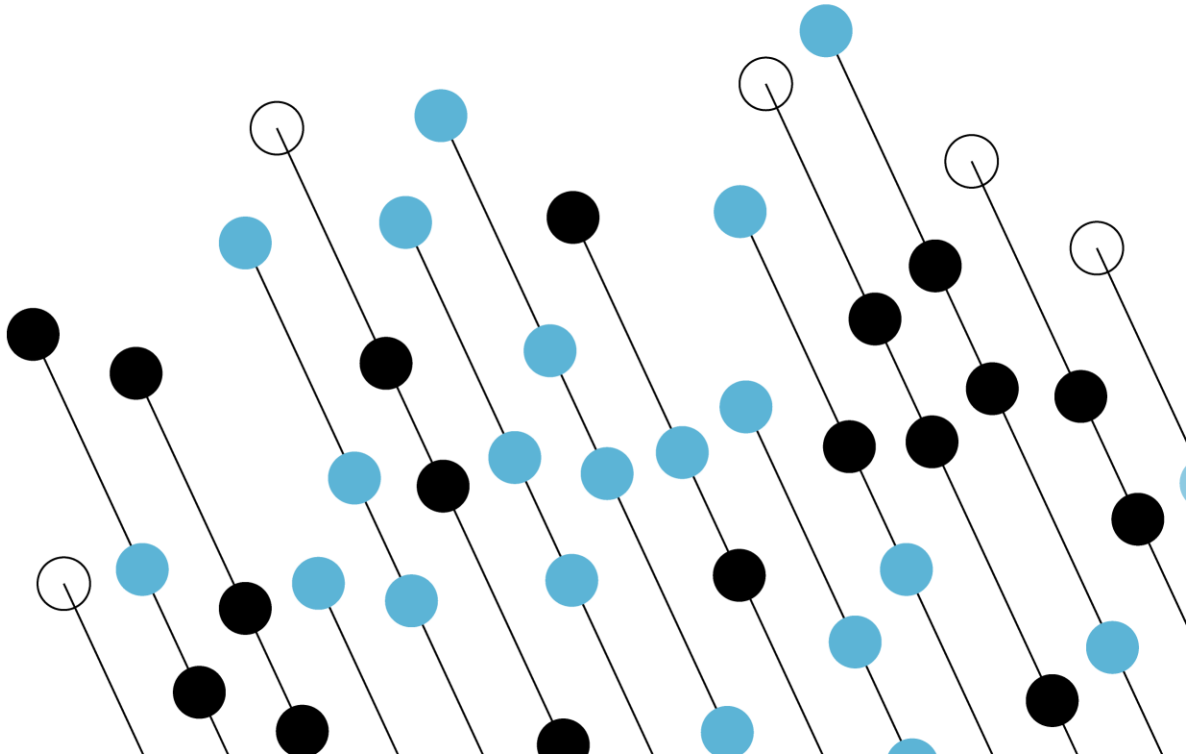
Core Cloud and Networking Skills

Part 1 - Day 2

QA

Learn. To Change.

Encryption



Encryption

Think of encryption as a secret code for your data. It transforms your plain, readable information into a jumbled mess that only someone with the right key can understand.

This key acts like a special password, allowing authorised users to unlock the code and access the original data.

Encryption safeguards your sensitive information, whether it's stored on your computer or traveling across the internet, ensuring only those with permission can see it.

Encryption: Key concepts

Plaintext: The original, unencrypted data or message that is readable.

Ciphertext: The encrypted, unreadable form of the data produced after encryption.

Encryption algorithm: A mathematical method or procedure used to convert plaintext into ciphertext.

Encryption key: A secret sequence of characters that the encryption algorithm uses to encrypt or decrypt data. The key must be kept confidential, as possession of it enables decryption.

Encryption: History

The history of encryption dates back thousands of years. Here are the key milestones in the early development of encryption:

- Ancient Egypt and the Caesar Cipher (c. 1900 BCE - 1st Century BCE)
- The Scytale Cipher (c. 500 BCE)
- The Vigenère Cipher (16th Century)
- Number Stations (World War I Era)
- The Enigma Machine (World War II Era)

Encryption: Number station

Number station is a form of secure communication for spies or intelligence operatives, allowing for the transmission of secret messages without the need for direct contact.

Messages were transmitted in plain text but encoded through methods like the **one-time pad.** They were highly secure and could be decoded only by those with the corresponding key.

Encryption: Number station

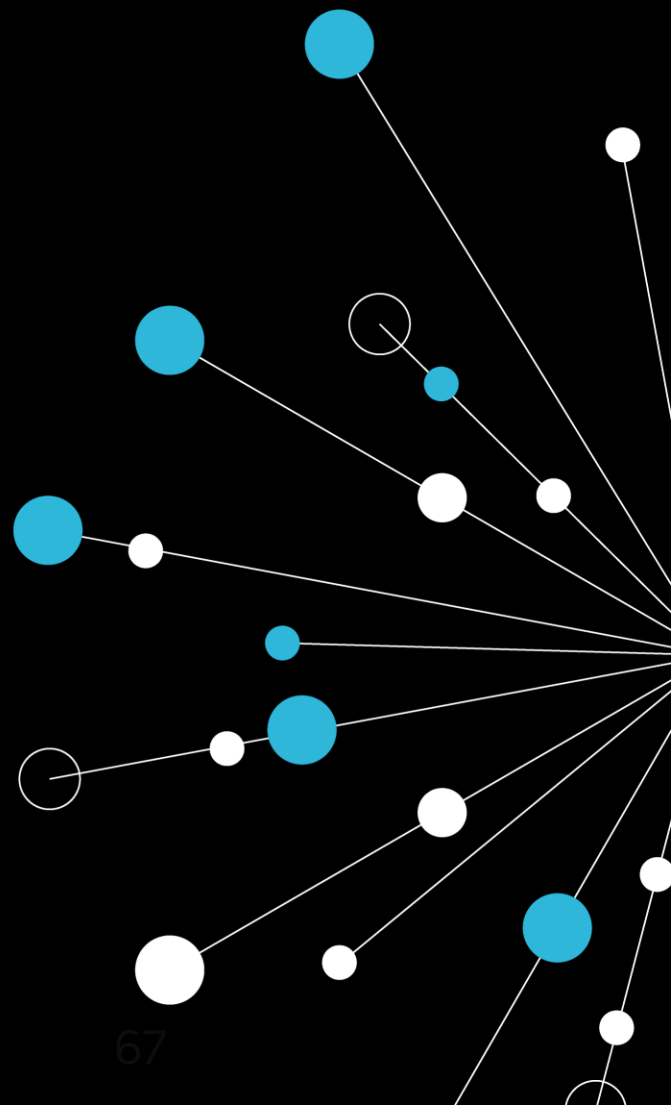
How it works:

- The recipient of the broadcast would have a **pre-shared key** (a random string of numbers or letters) that they use to decrypt the message.
- Each number or sequence from the broadcast corresponds to a letter, and the key is applied to decode the message.
- Because the key is random and used only once, this method is considered unbreakable if properly executed.

Activity 8: OTP

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Encryption: Uses

Data protection: Ensures the security of sensitive information such as personal details, passwords, and financial data.

Secure communication: Maintains the privacy of messages sent via email, messaging apps, or online platforms.

File and disk encryption: Encrypts files or entire drives to prevent unauthorised access in case a device is lost or stolen.

Virtual private networks (VPNs): Secures internet traffic to protect online activities and prevent eavesdropping.

Digital signatures: Confirms the authenticity and integrity of messages or documents.

Encryption: Types

Symmetric encryption uses the same key for both the encryption and decryption processes.

Asymmetric encryption involves a pair of keys: a public key used for encryption and a private key used for decryption.

Hashing is a one-way process that converts data into a fixed-size value, known as a hash.

Symmetric encryption

Symmetric encryption uses the same key for both locking (encrypting) and unlocking (decrypting) data.

This key must be kept confidential and shared only between authorised parties.

Think of it as a secret code known only to those who need to access the information, ensuring secure exchange and protection of sensitive data.

Symmetric encryption

Advantages:

- Symmetric encryption is fast, making it perfect for encrypting large amounts of data quickly.
- It's like a simple lock and key, straightforward and efficient for both encryption and decryption.

Disadvantages:

- The shared key is like a master password. If it falls into the wrong hands, the entire system is compromised.
- Managing keys becomes a challenge in large systems with many users, as each pair needs a unique shared key.

Asymmetric encryption

Asymmetric encryption uses a pair of keys (public and private), generated using cryptographic algorithms.

The sender uses the recipient's public key to encrypt the data.

The recipient uses their private key to decrypt the data encrypted with their public key.

Since the private key is secret, only the intended recipient can decrypt and read the message.

Asymmetric encryption

Advantages:

- Public keys can be freely shared without compromising private keys.
- Unlike symmetric encryption, there is no requirement to securely transmit keys.

Disadvantages:

- Asymmetric encryption is computationally intensive, making it slower compared to symmetric encryption.
- Managing key pairs (generation, storage, and distribution) adds additional overhead.

Hashing

Hashing is a one-way cryptographic technique used to generate a fixed-length representation of data.

It plays a critical role in verifying data integrity, securing passwords, and supporting technologies like blockchain.

Hashing cannot be reversed, making it ideal for scenarios where data needs to be verified but not decrypted.

Hashing

If we hash the word ‘hello’ using SHA-256, the output is:

2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824

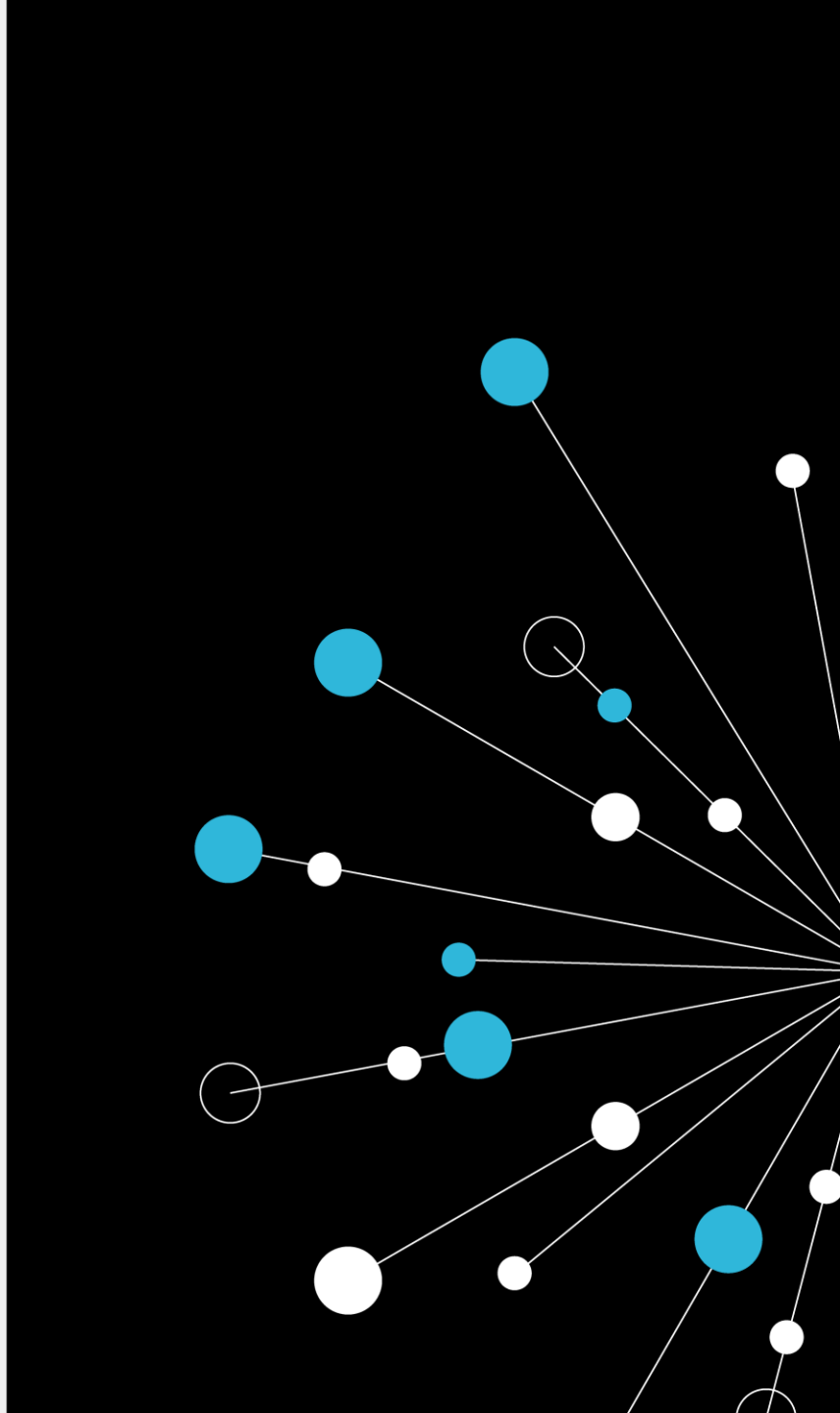
If we change the input slightly to ‘Hello’ (uppercase H), the resulting hash is completely different:

185f8db32271fe25f561a6fc938b2e264306ec304eda518007d1764826381969

Activity 9: Encryption Q&A

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Encryption at rest

Full disk encryption (FDE): Encrypting an entire disk to ensure all files and data on it are protected. Prevents unauthorised access if the device is stolen.

File-level encryption: Encrypting specific files or data rather than an entire disk. Ensuring that backup data is secure and cannot be read if stolen.

Database encryption: Encrypting data within databases to protect sensitive information. Encrypting files or databases stored in cloud or on-site environments to protect sensitive data.

File level encryption:

EFS

7-Zip

GPG

WinRAR

Drive encryption:

BitLocker

VeraCrypt

Tomb

Encryption in transit (online)

Web browsing: Using protocols like HTTPS (Hypertext Transfer Protocol Secure) to secure communication between a web browser and a website, protecting user data from eavesdropping or man-in-the-middle (MITM) attacks.

Email encryption: Ensuring that emails and attachments are securely transmitted between email servers and recipients.

VPNs (Virtual Private Networks): Securing online activity and traffic between a user and a remote network by encrypting data transmitted over public networks.

File transfers: Encrypting files during transfers to prevent interception, such as with SFTP (Secure File Transfer Protocol).

Encryption: Digital signatures

To verify a digitally signed message or document, the recipient follows these steps:

1. Using the sender's public key, the recipient decrypts the digital signature.
2. The recipient then calculates a unique fingerprint (hash) of the received message or document.
3. The recipient compares the decrypted digital signature with the newly calculated fingerprint. If they match, it means the message hasn't been tampered with and is truly from the sender.

This ensures that the message is authentic and hasn't been altered in transit.

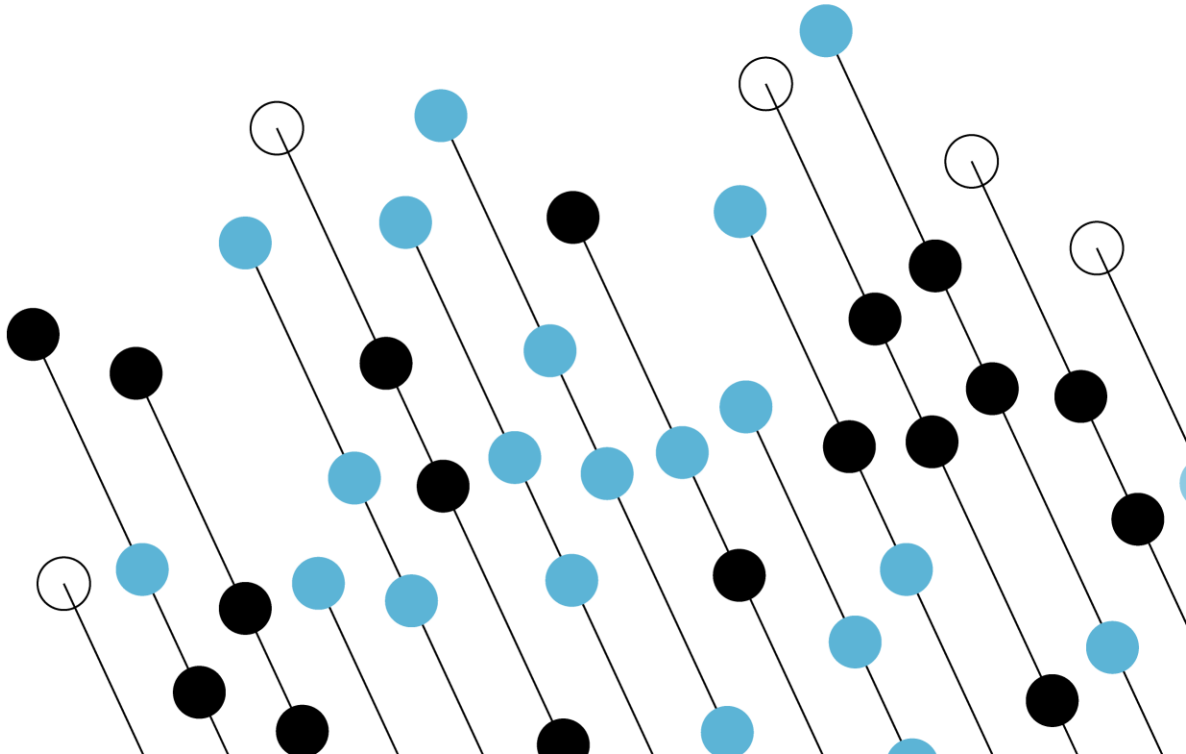
In summary: Encryption

Encryption is a cornerstone of modern digital security, safeguarding sensitive data and communications from unauthorised access.

By transforming readable information into an unreadable format using cryptographic algorithms and keys, encryption ensures confidentiality, integrity, and authenticity.

Whether protecting stored data (encryption at rest) or securing information during transmission (encryption in transit), it plays a vital role in preventing data breaches, ensuring privacy, and building trust in digital systems.

What is the cloud?



What is the cloud?

Think of the cloud as a vast, digital landscape where you can store, manage, and access your data and applications without the limitations of your own computer.

Instead of relying on your device's hard drive or a physical server in your office, you can leverage the power of remote servers connected through the internet.

This network of servers, managed by third-party cloud service providers, offers a wide range of computing resources, including storage, processing power, and software applications, all available on demand.

What is the cloud?

The cloud is...

- Hardware
- Networks
- Storage
- Services
- Interfaces

Cloud services deliver...

- Software
- Infrastructure
- Storage

Cloud service providers are responsible for...

- Physical assets
- Maintenance

What is the cloud?

The National Institute of Standard and Technology (NIST) defines **five essential characteristics** of cloud computing:

1. On-demand self-service
2. Broad network access
3. Resource pooling
4. Rapid elasticity
5. Measure service

What is the cloud?

On-demand self-service:

Customers can provision services without direct interaction with the provider, using dashboards and APIs. Providers like AWS, Azure, and Google Cloud offer this flexibility.

Broad network access:

Cloud services are accessible via standard network protocols (e.g., HTTP, TCP) on various devices like phones and laptops, though not necessarily over the public internet.

What is the cloud?

Resource pooling:

Cloud providers use shared resources to serve multiple tenants, dynamically allocating them as needed.

Customers don’t need to know the physical location of resources, though general locations may be specified for regulatory or performance reasons.

What is the cloud?

Rapid elasticity:

Cloud resources can be quickly scaled up or down based on demand, allowing users to adjust resources efficiently. This scaling is often automatic and transparent to the customer.

Measured service:

Cloud resources are typically provided on a pay-per-use basis, and customers can monitor usage in real-time to manage costs and performance.

The big three cloud service providers

Amazon Web Services (AWS)

The largest and most widely used cloud platform globally.

Key services:
EC2 (Elastic Compute Cloud), S3 (Simple Storage Service), RDS (Relational Database Service), Lambda, and others.

Microsoft Azure

Widely used by businesses, especially for its integration with Microsoft software.

Key services:
Azure Virtual Machines, Azure Blob Storage, Azure SQL Database, Azure Active Directory, and more.

Google Cloud Platform (GCP)

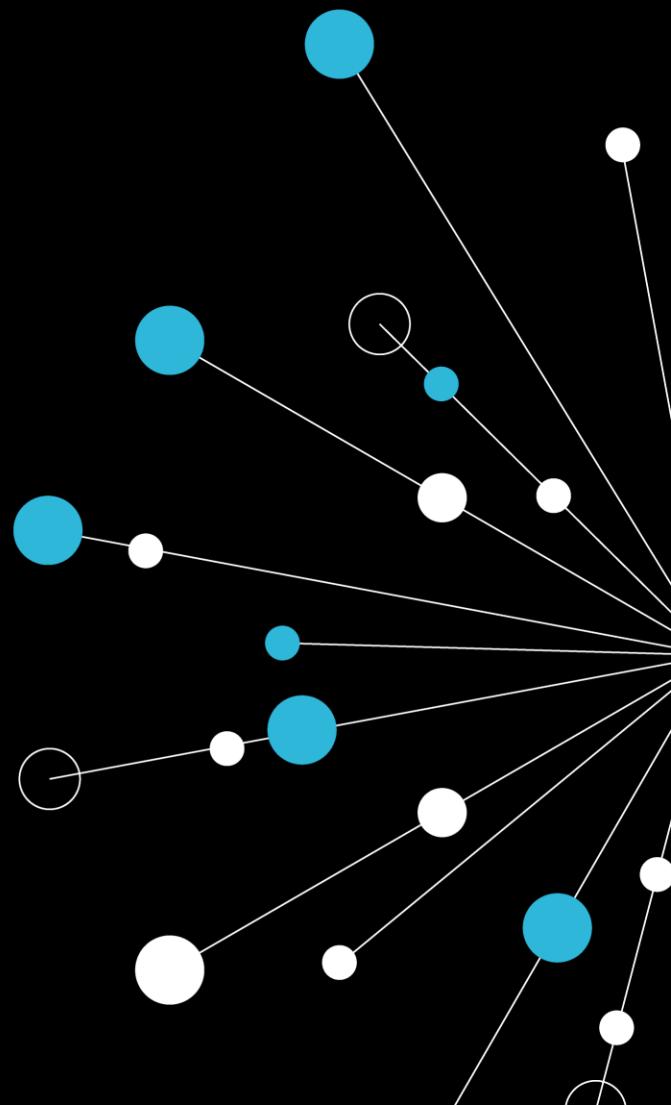
Popular for its AI and machine learning tools.

Key services:
Google Compute Engine, Google Cloud Storage, BigQuery, Google Kubernetes Engine (GKE), and others

Activity 10: Cloud models

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



What is the cloud?

The NIST (National Institute of Standards and Technology) definition of **cloud computing** outlines three main **service models**:

1. Infrastructure as a Service (IaaS)
2. Platform as a Service (PaaS)
3. Software as a Service (SaaS)

Cloud service models

On-premises	IaaS	PaaS	SaaS
Application	Application	Application	Application
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
Operating system	Operating system	Operating system	Operating System
Virtualisation	Virtualisation	Virtualisation	Virtualisation
Physical server	Physical server	Physical server	Physical server
Physical storage	Physical storage	Physical storage	Physical storage
Physical network	Physical network	Physical network	Physical network

Managed by customer

Managed by vendor

Infrastructure as a Service (IaaS)

Consumers can provision fundamental computing resources such as virtual machines, storage, and network capacity directly from the cloud provider.

The consumer has control over the operating systems, deployed applications, and storage but relies on the provider to manage the underlying physical infrastructure.

This is ideal for businesses or developers that require flexibility to run arbitrary applications, test software, or build complex IT environments without investing in physical hardware.

Examples:

- Amazon EC2 (Elastic Compute Cloud)
- Google Compute Engine
- Microsoft Azure virtual machines

Platform as a Service (PaaS)

Consumers can deploy custom applications created using programming languages, libraries, and tools supported by the provider.

The cloud provider manages the underlying hosting environment, including the operating system, servers, and infrastructure.

Consumers focus solely on building, deploying, and maintaining their applications without worrying about infrastructure management or scaling concerns.

This is well-suited for developers looking for an environment to develop and deploy applications without managing hardware or software updates.

Examples:

- Google App Engine
- Microsoft Azure App Service
- Heroku

Software as a Service (SaaS)

Consumers have access to fully functional, ready-to-use software applications hosted on the cloud.

Consumers use applications directly via a web browser or an app without any responsibility for managing the underlying infrastructure, operating system, or application deployment.

Users can configure some application settings but not core functionality or backend operations.

This is best for end-users who need easy access to software solutions without the complexities of installation, updates, or hardware requirements.

Examples:

- Google Workspace (Docs, Gmail, Drive)
- Microsoft 365
- Salesforce

Cloud consumer services

The NIST Cloud Computing Roadmap describes ‘services available to a cloud consumer’ as the resources and functionalities provided by cloud service providers to meet different needs across the three models.

- **IaaS consumer services:** Backup and recovery, CDN, storage, services management, platform hosting, compute
- **PaaS consumer services:** Development and testing, business intelligence, database, application deployment, integration
- **SaaS consumer services:** Document management, collaboration, DRM, sale, billing, ERP, human resource, social networks, financials, content management, email and office productivity

Other types of cloud service model

XaaS (Anything as a Service): A broad term that refers to the delivery of various services and solutions over the internet, leveraging cloud computing. This includes IaaS, PaaS, and SaaS, as well as more specialised offerings.

Backup as a Service (BaaS): Provides cloud-based data backup and recovery solutions. Examples include Veeam Backup and Carbonite.

Desktop as a Service (DaaS): Offers virtual desktops that can be accessed remotely over the internet. Examples include Citrix Virtual Apps and Desktops and Amazon WorkSpaces.

Other types of cloud service model

Communication as a Service (CaaS): Cloud-based communication services such as VoIP, video conferencing, and messaging. Examples include Zoom and Twilio.

Gaming as a Service (GaaS): Offers cloud-based gaming experiences, enabling users to stream and play without high-end hardware. Examples include Google Stadia and NVIDIA GeForce NOW.

AI as a Service (AIaaS): Provides AI capabilities like machine learning, NLP, and computer vision as on-demand services. Examples include IBM Watson and Google AI.

Cloud deployment models

This refers to the strategies and frameworks used to configure, deploy, and manage cloud computing environments based on specific organisational needs.

They define how resources like storage, computing power, and applications are hosted and accessed.

The key deployment models are:

- **private cloud**.
- **public cloud**.
- **hybrid cloud**.
- **community cloud**.

Private cloud

A private cloud is dedicated exclusively to a single organisation. Resources are not shared with other entities, ensuring greater security, control, and customisation.

It can be hosted on-premises or managed by a third-party provider.

They are commonly used by organisations that handle sensitive data or require strict compliance, such as government agencies or financial institutions.

Examples:

- Microsoft Azure Stack
- VMware vSphere
- OpenStack

Public cloud

A public cloud is where resources are owned, managed, and maintained by a third-party cloud service provider and shared across multiple customers.

Services are delivered over the internet, offering scalability and cost efficiency.

This is ideal for businesses with fluctuating demands and those looking for flexibility without managing infrastructure.

Examples:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud

Hybrid cloud

A hybrid cloud combines private and public cloud models, allowing data and applications to move between them.

Organisations can keep sensitive operations on a private cloud while leveraging the public cloud for less critical workloads or to scale as needed.

They provide flexibility, cost optimisation, and a balance between security and scalability.

Examples:

- AWS Outposts.
- Microsoft Azure Arc
- Google Anthos

Community cloud

A community cloud is shared by multiple organisations with similar goals, regulatory requirements, or needs.

They are often seen in sectors like healthcare or education, where participants benefit from a shared infrastructure tailored to their specific use case.

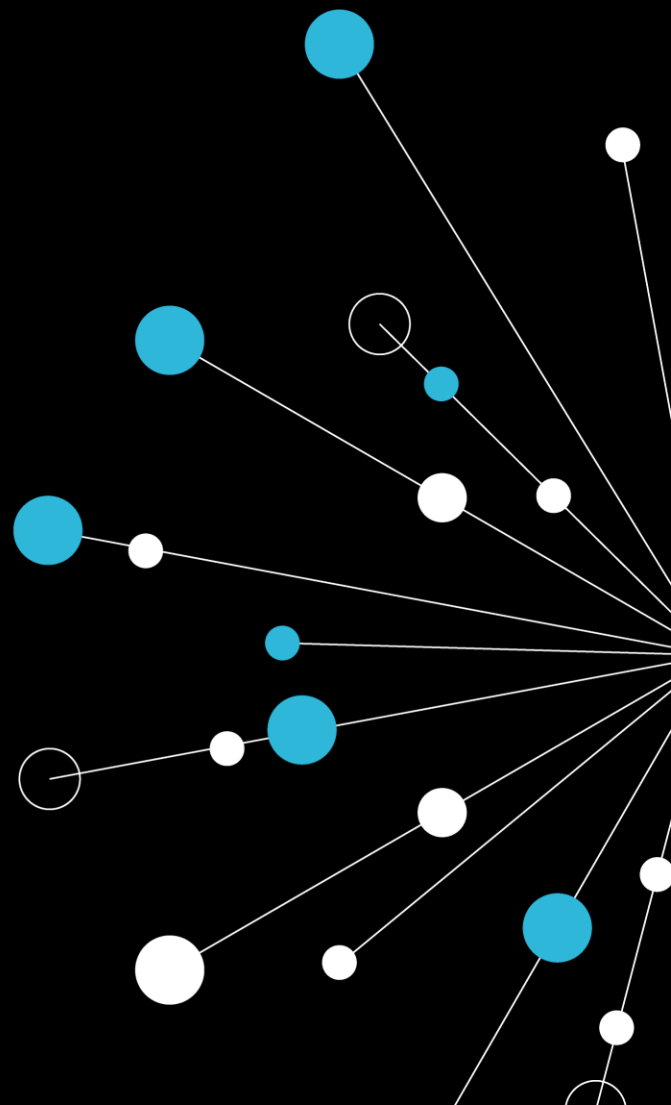
Examples:

- CERN OpenStack
- UKCloud
- Health Information Exchange (HIE)

Activity 12: Cloud Q&A

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: What is the cloud?

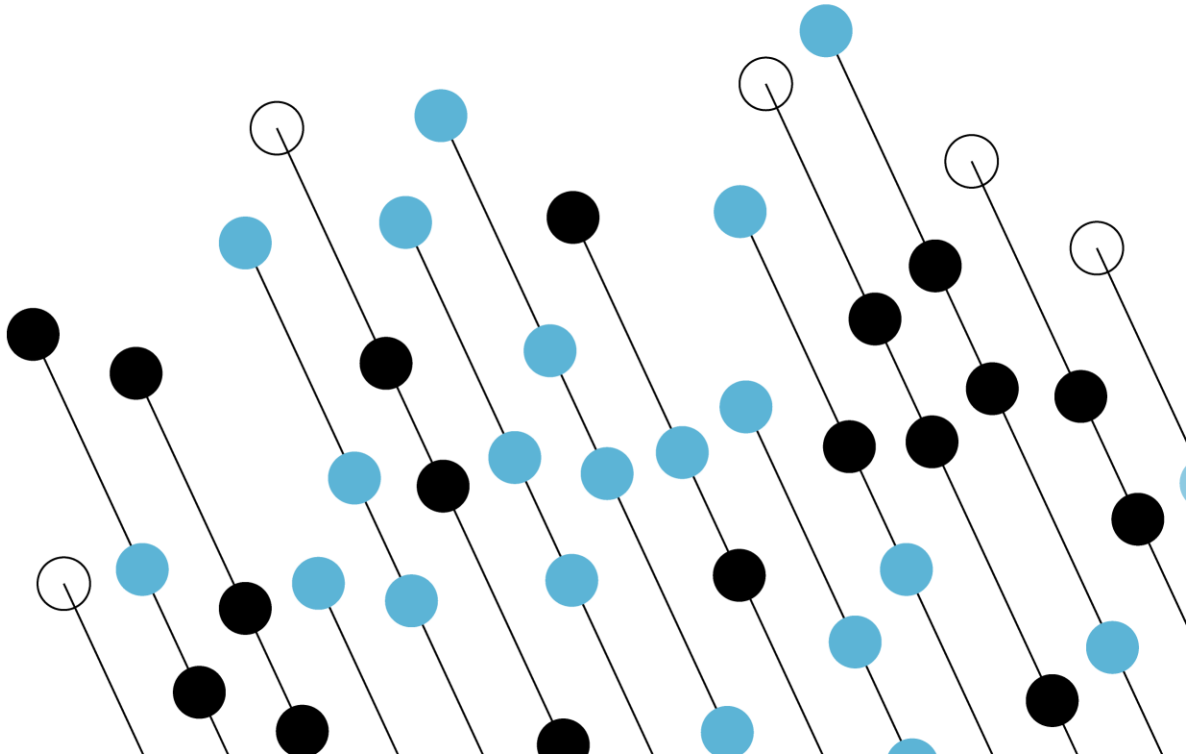
Cloud technology provides flexible, scalable, and cost-effective solutions for delivering computing resources.

It is defined by three main service models: IaaS (Infrastructure as a Service) for virtualised hardware, PaaS (Platform as a Service) for application development, and SaaS (Software as a Service) for ready-to-use software applications.

Additionally, cloud technology is deployed through four deployment models: private (dedicated to one organisation), public (shared resources over the internet), hybrid (a mix of private and public), and community (shared by organisations with common needs).

Together, these models offer organisations the ability to innovate, scale, and securely manage their IT resources.

Virtualisation



Virtualisation

Virtualisation is the process of creating a virtual version of computing resources, such as servers, storage devices, networks, or operating systems, instead of relying on physical hardware.

It uses software called a hypervisor to enable multiple virtual machines (VMs) to run on a single physical machine, each with its own operating system and applications.

Reasons for using virtualisation

Resource optimisation: Virtualisation allows better utilisation of physical hardware by running multiple virtual machines on a single server, reducing idle resources.

Cost efficiency: Reduces hardware costs as fewer physical machines are needed. This also minimises power, cooling, and maintenance expenses.

Scalability: Virtual environments can quickly scale resources up or down to meet changing business demands without the need for new physical infrastructure.

Reasons for using virtualisation

Testing and development: Virtual machines allow developers to test software or applications across multiple environments without requiring dedicated hardware.

Simplified management: Centralised management tools make it easier to monitor and control virtual resources, reducing administrative overhead.

Flexibility and agility: Enables rapid deployment and migration of virtual machines, improving responsiveness to business needs.

Hypervisors

A **hypervisor** is software or firmware that creates and manages virtual machines (VMs) on a physical host machine.

It allows multiple operating systems (OS) to run simultaneously on a single physical machine by allocating resources like CPU, memory, and storage to each virtual machine.

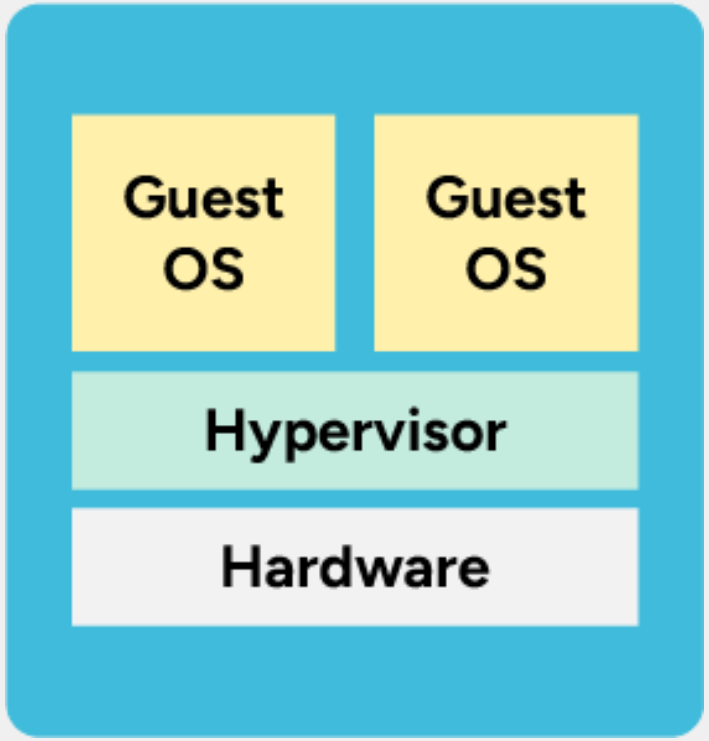
The hypervisor ensures isolation between VMs, so they operate independently of one another.

Type 1 hypervisor (bare-metal)

A type 1 hypervisor runs directly on physical hardware without an underlying operating system.

Examples:

- VMware ESXi
- Citrix Hypervisor (formerly XenServer)
- KVM (Kernel-based Virtual Machine)

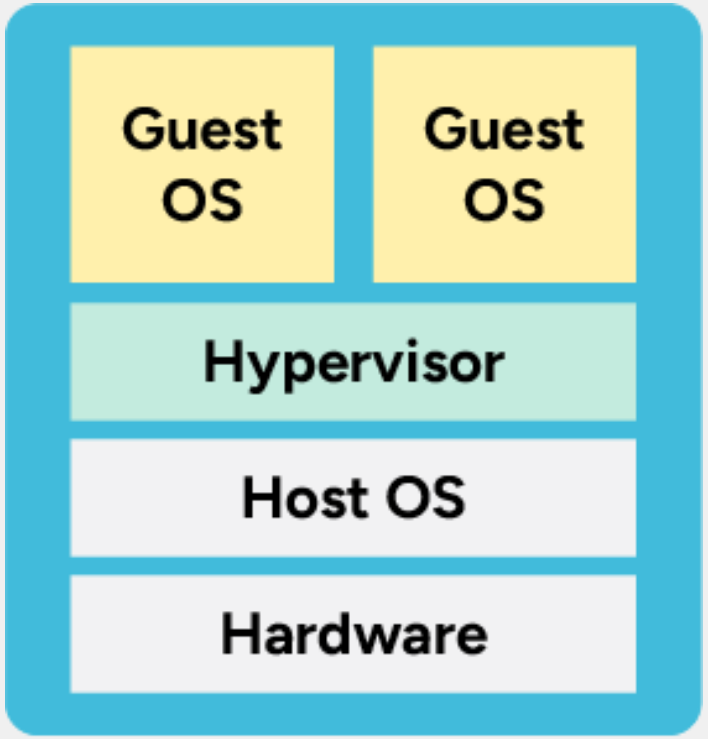


Type 2 hypervisor (hosted)

A type 2 hypervisor runs on top of an existing operating system, like an application.

Examples:

- Oracle VirtualBox
- VMware Workstation
- Parallels Desktop
- QEMU



Virtualisation: Factors to consider for resource sharing

- **CPU:** Assign sufficient CPU cores based on the workload. Avoid over-provisioning, as it can degrade performance on the host machine.
- **Memory (RAM):** Allocate enough RAM for the VM to run smoothly without impacting the host system. Plan for both OS and application requirements.
- **Storage:** Choose between dynamic (thin) or fixed (thick) disk space. Fixed storage is pre-allocated, while dynamic grows as needed.
- **Network bandwidth:** Ensure adequate network resources for the VM to support communication, especially for servers or applications requiring high throughput.

Types of virtualisation

CPU hardware extensions in virtualisation are built-in features in modern processors that improve VM performance and efficiency.

They offload virtualisation tasks to the CPU, enhancing isolation, management, and execution, while reducing overhead in multi-VM environments.

Examples:

- Intel VT-X
- AMD-V
- ARM Virtualisation Extensions

Application virtualisation enables applications to run on a device without being installed on its operating system.

The app is encapsulated in a virtual environment, allowing remote access or streaming over a network, offering flexibility and better resource management.

Examples:

- VMWare ThinApp
- Microsoft App-V
- Citrix XenApp

Types of virtualisation

Storage virtualisation abstracts and consolidates physical storage from multiple devices into a single, unified virtual storage system, appearing as one resource to users and applications, despite being spread across different physical devices.

Examples:

- Amazon NooBaa
- Azure Blob Storage
- Storage Area Networks (SANs)

Virtual Desktop Infrastructure (VDI) hosts desktop environments on a centralised server, enabling remote access to desktop operating systems and applications over a network, instead of running them locally on a personal device.

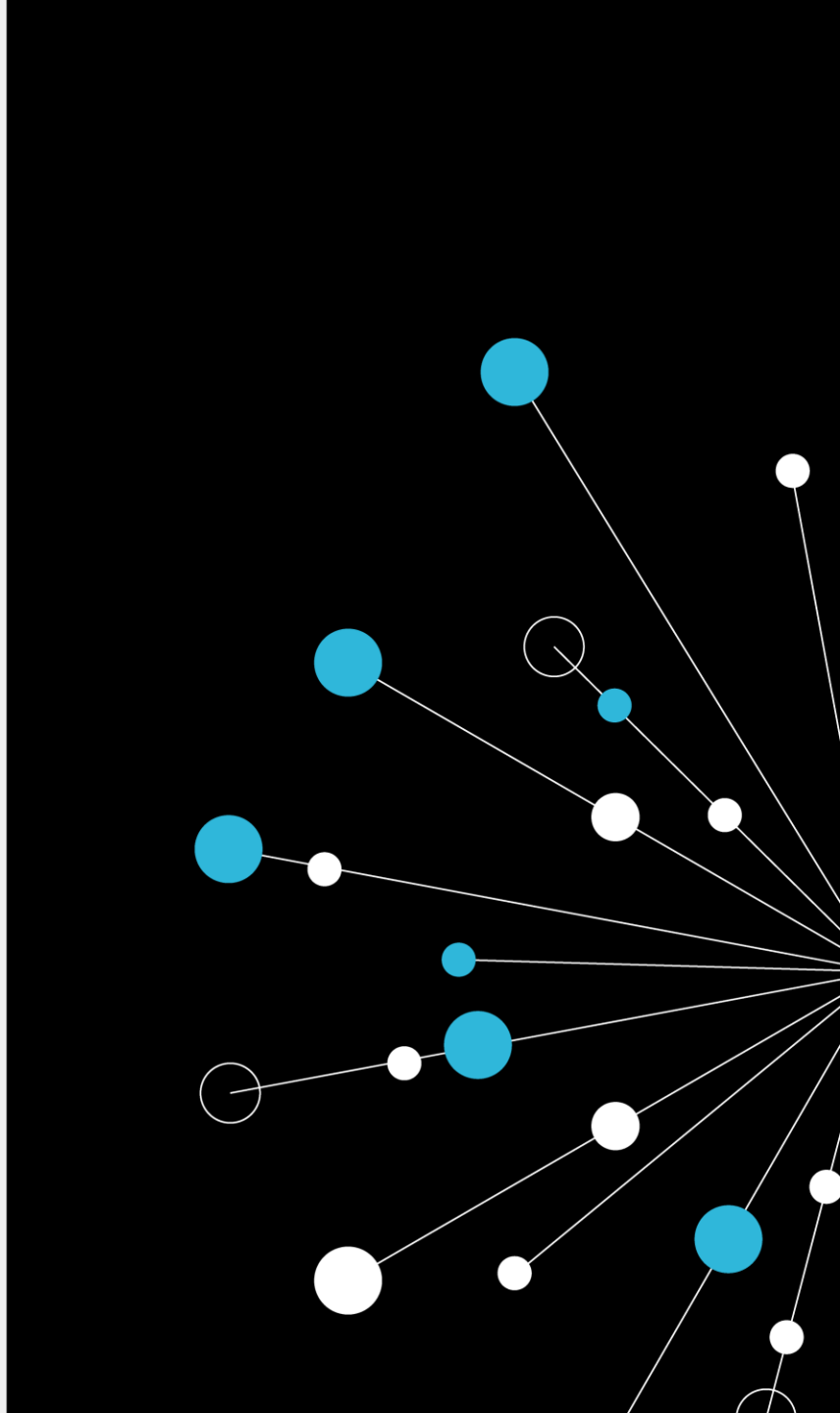
Examples:

- Amazon Workspaces
- VMWare Horizon 7
- Microsoft VDI

Activity 12: Virtualisation Q&A

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



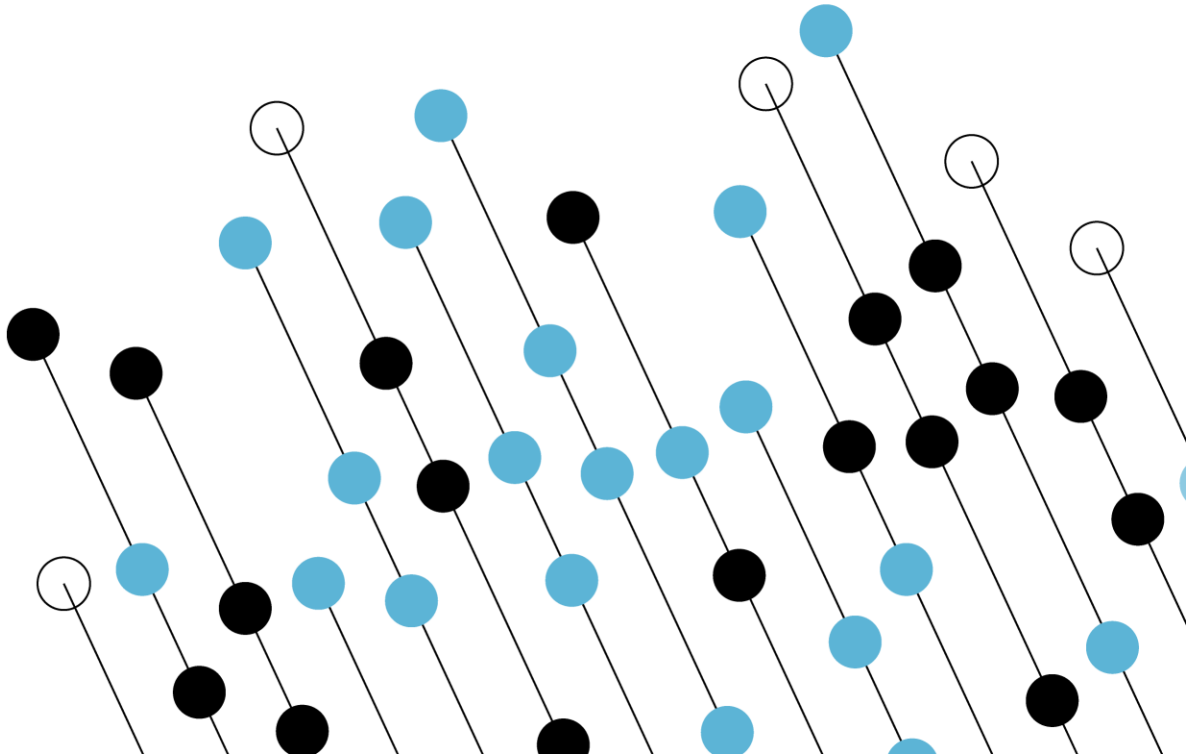
In summary: Virtualisation

Virtualisation is a technology that allows for the creation of virtual instances of physical resources, such as servers, storage, and desktops.

It enhances resource efficiency, flexibility, and scalability while simplifying management and reducing costs.

Virtualisation enables organisations to run multiple systems on a single physical machine, improving performance, security, and the ability to adapt to changing demands.

Backup and storage



Backup and storage

Storage refers to the systems and devices used to save and retrieve data. Modern storage solutions are designed to meet varying performance, capacity, and accessibility needs.

Backups are duplicate copies of data that are stored in a separate location from the original data.

The primary purpose of backups is to ensure that, in the event of data loss, critical information can be recovered quickly and effectively, minimising downtime and preventing permanent loss of important data.

While creating backups is crucial for protecting data, the ability to restore that data when needed is equally important.

Types of storage

Local storage: Data is stored on physical devices like hard drives (HDDs), solid-state drives (SSDs), or Network Attached Storage (NAS). It offers fast access but can be vulnerable to physical damage.

Network storage: Data is stored on servers within a network and can be accessed remotely by authorised users, providing shared storage across devices.

Cloud storage: Data is stored remotely on servers managed by a cloud provider, offering scalability, remote access, and redundancy.

Object storage: A method used in cloud environments where data is stored as objects, rather than files or blocks, offering flexibility and efficient data retrieval.

Data backup

Data backup involves creating copies of specific files, folders, or databases to protect valuable data from loss or corruption.

It focuses on backing up individual pieces of information, such as documents, photos, videos, or important records.

The primary goal is to ensure that important files can be recovered in case of accidental deletion, data corruption, hardware failure, or other disasters.

Typically, recovery is faster, but only for data files.

This is ideal for personal users or businesses that need to recover individual files.

System backup

System backup (often called a disk image backup or system image backup) involves creating a complete copy of the entire system, including the operating system, installed applications, system settings, and user data.

The goal of a system backup is to restore the entire system to a previous working state, including both the operating system and all the installed software, in case of a failure or disaster.

It can be slower since it involves restoring the full system, but it ensures the system is fully operational as it was before the failure.

This is best for situations where the entire system needs to be recovered due to a crash, corruption, or failure.

The 3-2-1 backup rule

The 3-2-1 backup rule is a strategy for data protection and minimising loss risks.

It involves keeping **three** total copies of your data: the original and two backups.

To reduce failure risks, the backups should be stored on **two** different media types, such as an external hard drive and cloud storage.

One of the backups should be stored offsite to protect against disasters like fire, theft, or flooding at the primary location.

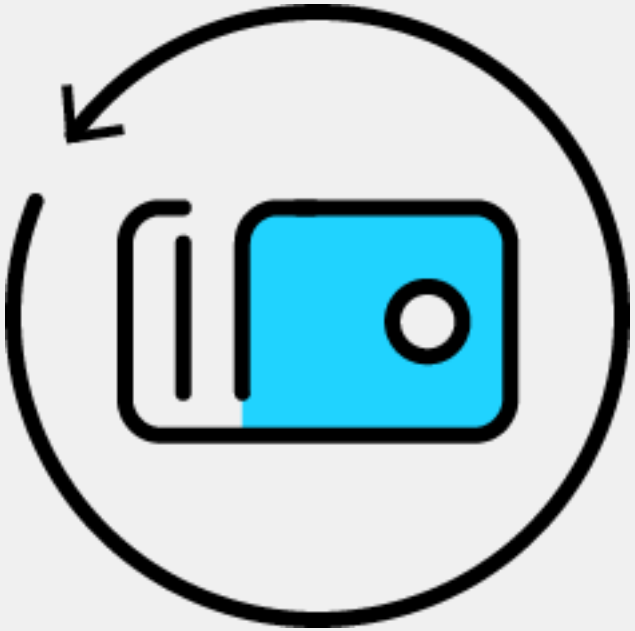
This ensures that data remains accessible even if one or two copies are lost.

Types of backup: Full backup

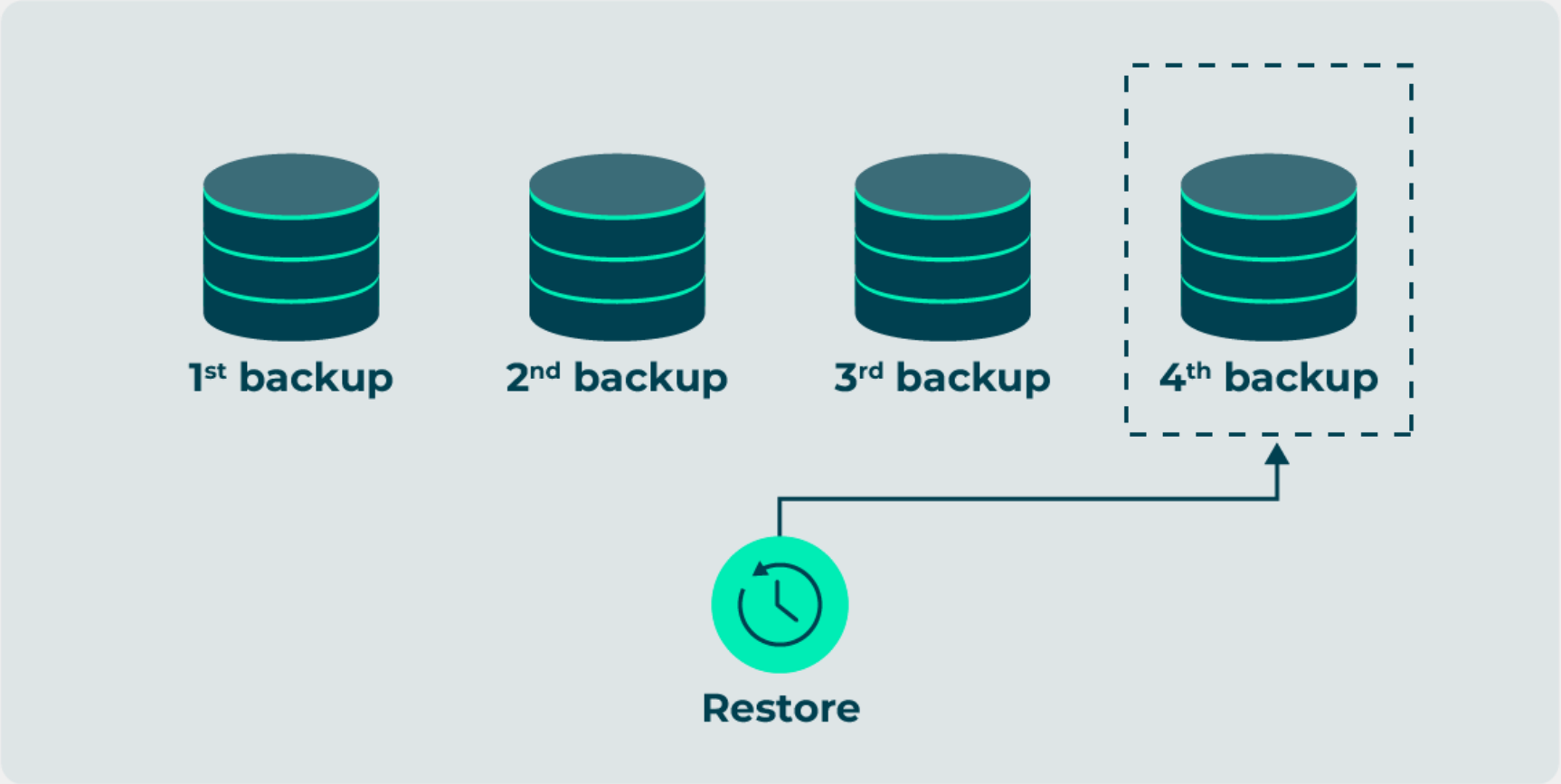
Full backup creates a complete copy of all data.

This type of backup is the easiest to manage and restore, as it contains everything required to recover the data in a single step.

However, it typically requires more storage space and takes longer to complete compared to other types of backups.



Full backup



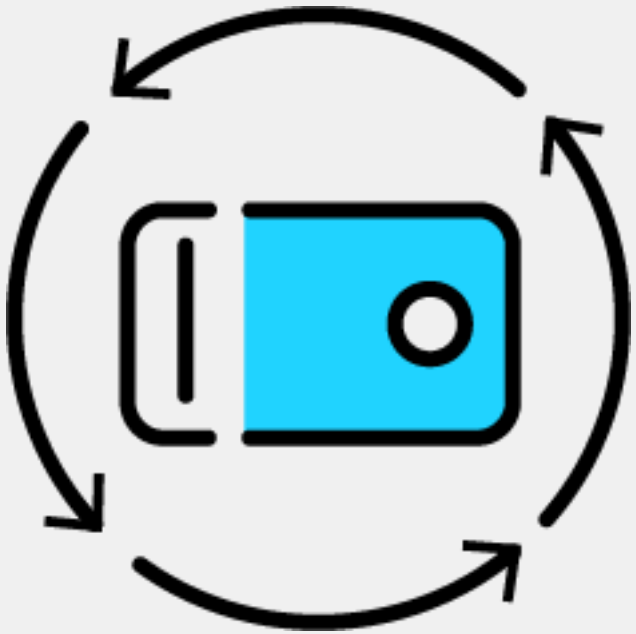
Types of backup: Incremental backup

Incremental backup only saves the changes made since the last backup (whether full or incremental).

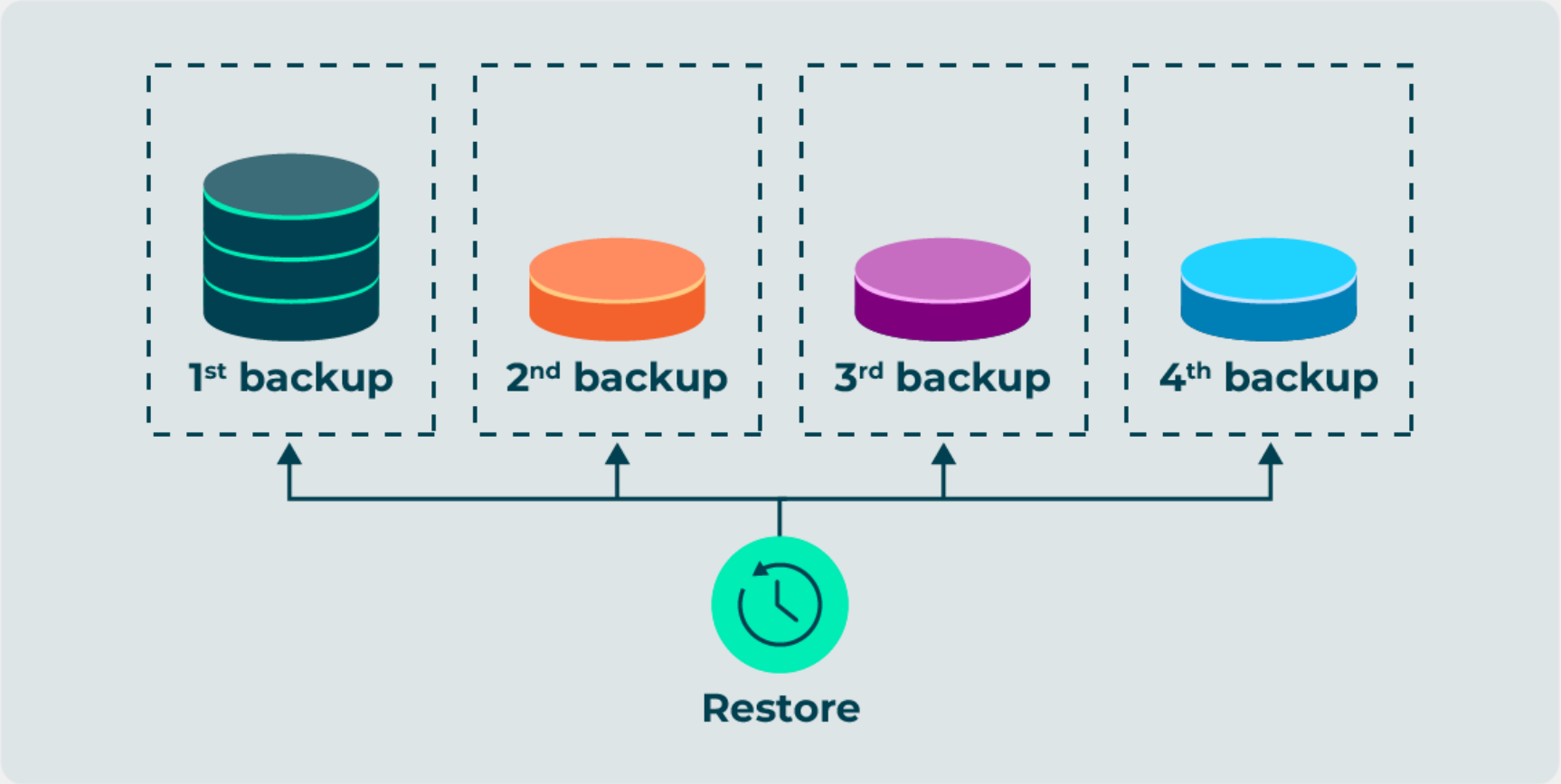
This method is quicker to perform and requires less storage space compared to full backups.

The downside is that restoration can be more complex.

It involves piecing together the last full backup with all subsequent incremental backups in sequence to fully recover the data.



Incremental backup



Types of backup: Differential backup

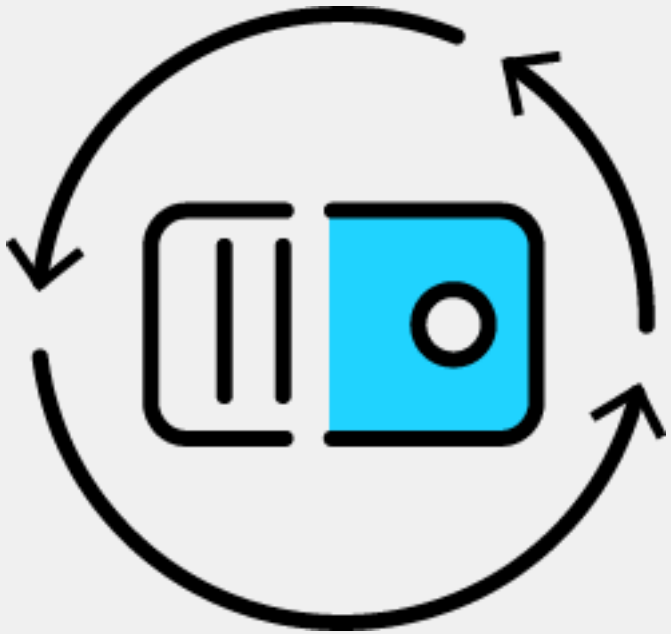
Differential backup saves changes made since the last full backup, but it does not rely on other differential backups.

It offers a balance between full and incremental backups.

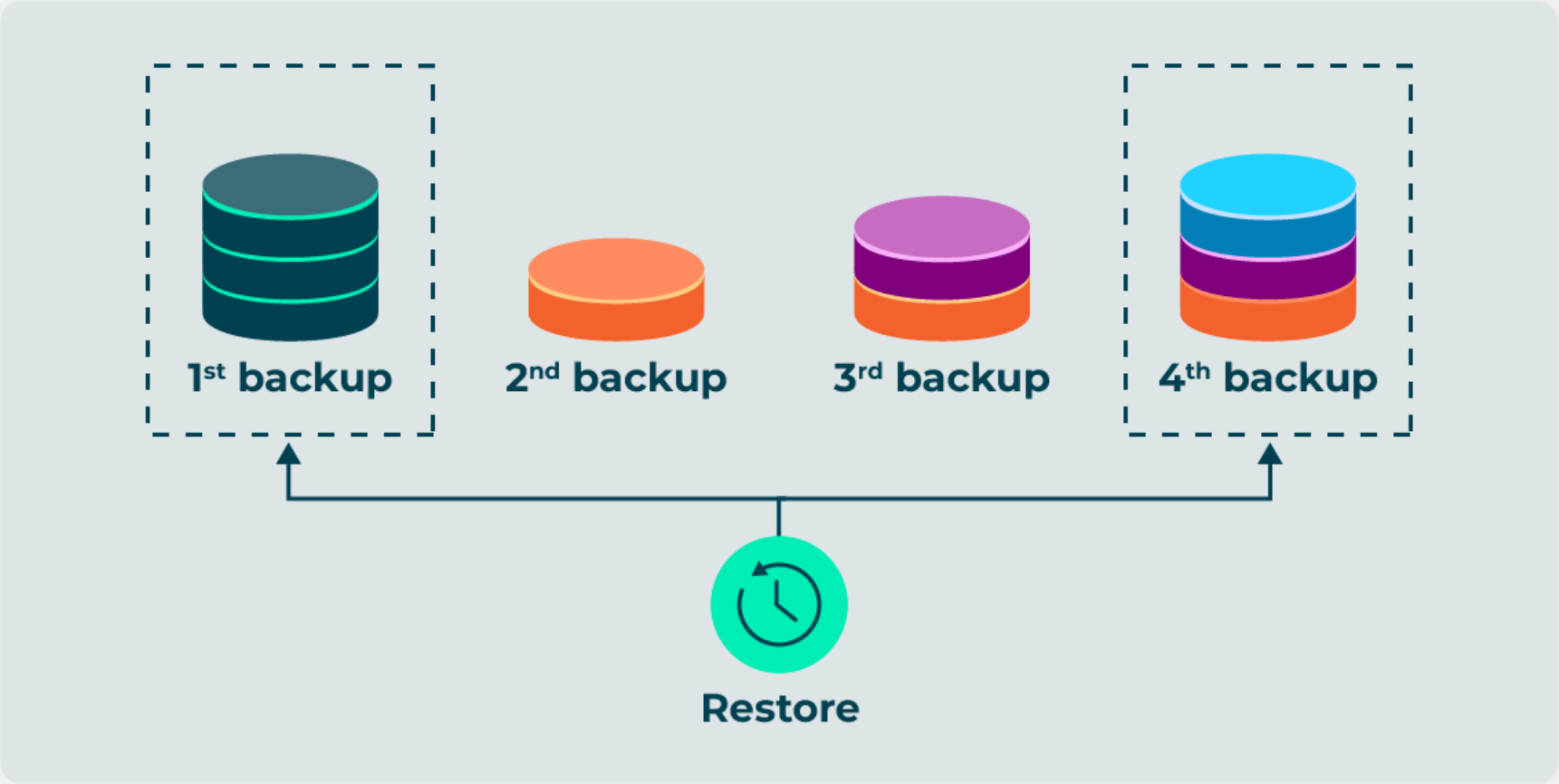
It’s quicker to perform than a full backup but easier to restore than an incremental backup.

During restoration, only the last full backup and the latest differential backup are needed.

This simplifies the recovery process compared to incremental backups.



Differential backup



Backup accessibility

Online backups refer to backups stored on cloud-based systems or servers that are accessible over the internet.

Data is immediately available for recovery or access. Backups are typically stored offsite (e.g., cloud storage services like Amazon S3, Google Cloud Storage).

It can be automated and run continuously or on a scheduled basis.

This is ideal for businesses or individuals who need frequent access to their backups and want to protect against local disasters (e.g., fires, theft, or hardware failure).

Backup accessibility

Offline backups are stored on physical media such as external hard drives, tapes, or DVDs, and are not connected to the internet or network.

Data is not accessible over the network unless the physical media is connected.

This is often used for long-term storage and protection from cyber threats like ransomware (since offline backups are not vulnerable to online attacks).

It is slower and more manual to access or restore.

Offline backups are ideal for businesses or individuals who want a backup that is physically isolated from their network, providing additional security against cyber threats.

Backup accessibility

Nearline backups are stored in a system that is connected to the network but not continuously accessible for active use.

They are typically stored on hard drives or other disk-based media that are ready to be connected to the system when needed.

They are used for backups that need to be quickly restored but are not required for constant access.

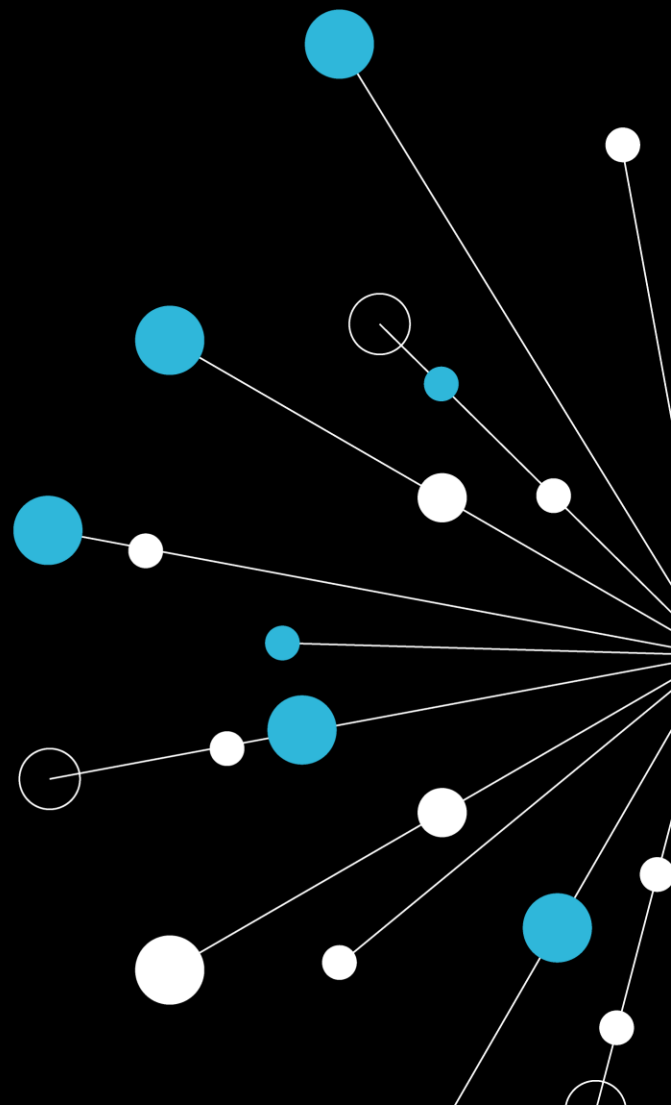
It is often less expensive than online backups, but they offer faster restoration than offline backups.

This is ideal for scenarios where the backup may not be accessed often but needs to be readily available for recovery in case of failure.

Activity 13: Backup and storage Q&A

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: Backup and storage

Backups and storage are critical components of data protection and recovery strategies.

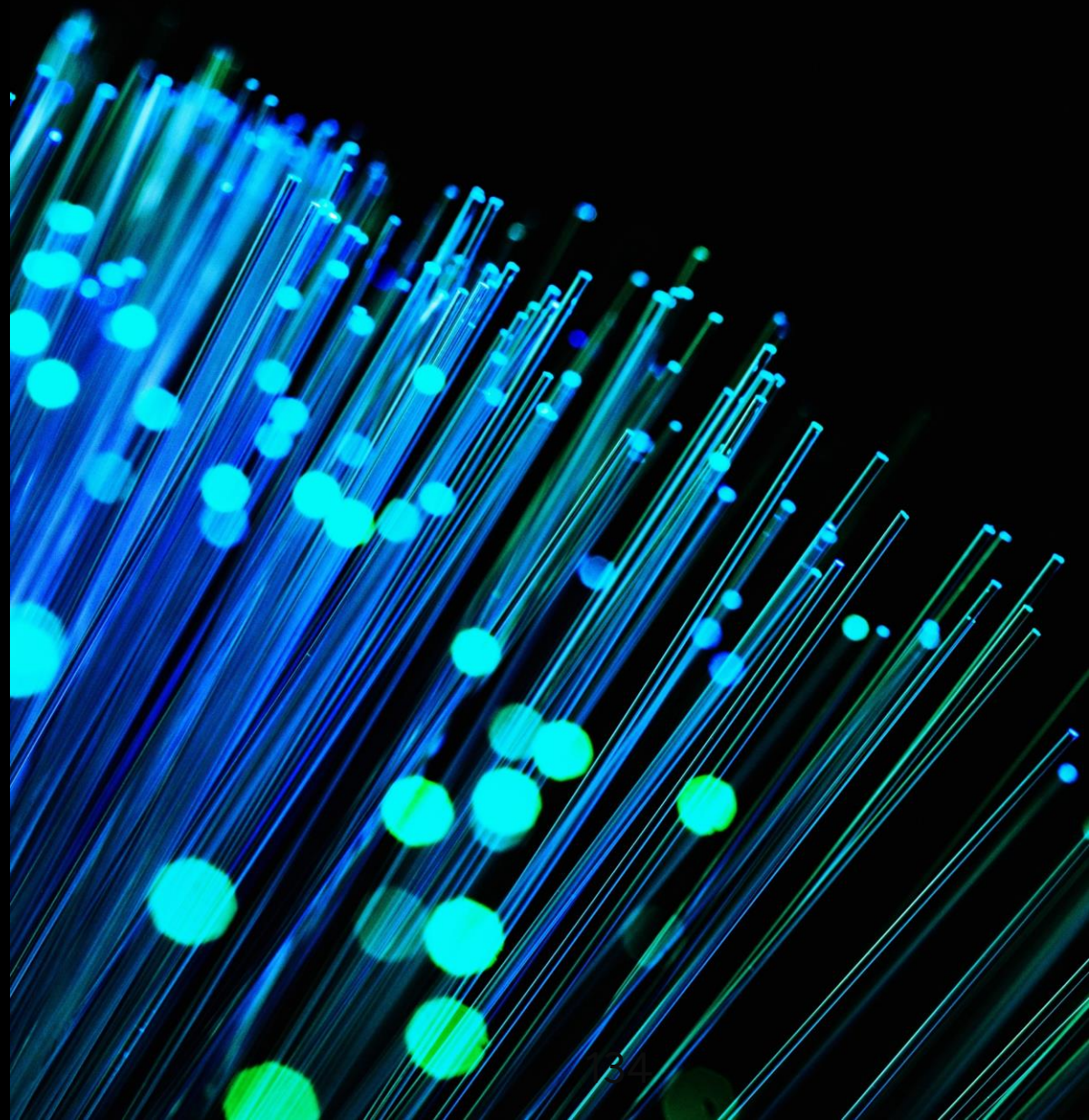
Backups ensure that data can be restored in case of loss or disaster, while storage solutions provide the infrastructure to securely house that data.

Effective backup strategies, such as the 3-2-1 rule, and the use of different backup types (full, incremental, and differential) ensure reliability and flexibility.

Combining efficient storage technologies like object storage and virtualisation enables scalable, accessible, and resilient data management, safeguarding critical information for both individuals and organisations.

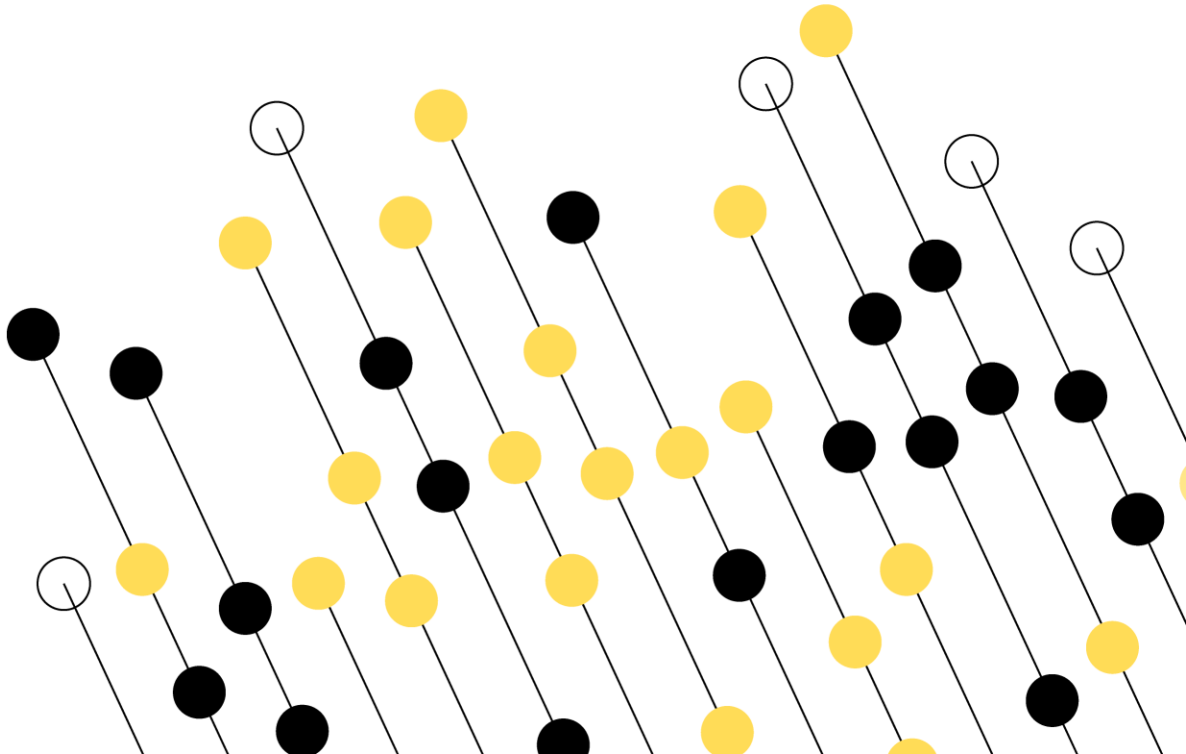
Core Cloud and Networking Skills

Day 3



Learn. To Change.

Wireless networking



Wireless networking

Wireless networking refers to the technology that allows devices to connect and communicate over a network without the need for physical cables.

It uses radio waves, infrared signals, or other wireless communication methods to transmit data between devices such as computers, smartphones, routers, and IoT devices.

Wireless networking

Common wireless networking technologies:

- **Wi-Fi:** For local wireless connections, widely used in homes, offices, and public hotspots.
- **Bluetooth:** For short-range wireless communication between devices like headphones, keyboards, and phones.
- **Cellular networks:** Used for mobile internet access via technologies like 4G and 5G.
- **Satellite communication:** For remote and global connectivity.

Wi-Fi

Wi-Fi, short for Wireless Fidelity, revolutionised how we connect to the internet. Its development spans decades, involving advancements in wireless communication and regulatory milestones. Here's an overview of its history:

1997: IEEE 802.11 Standard: The IEEE released the first Wi-Fi standard, enabling data rates up to 2 Mbps.

1999: Wi-Fi Alliance: Formed to promote Wi-Fi interoperability. The term 'Wi-Fi' was coined, and the 802.11b standard (11 Mbps) made Wi-Fi more practical.

2000s: Wi-Fi adoption surged with the 802.11g standard (54 Mbps). It became mainstream in homes, offices, and public spaces, integrated into laptops, smartphones, and gaming consoles.

2010s: High-speed standards like 802.11n and 802.11ac improved speed and reliability, powering streaming, IoT, and global Wi-Fi hotspots.

2020s: Wi-Fi 6 (802.11ax) enhanced crowded-network performance. Wi-Fi 7 promises faster speeds, lower latency, and better 5G integration, supporting smart homes and future technologies.

Wireless networking

Name	Speed	Frequency	Features
802.11a	54 Mbps	5 GHz	OFDM waveform
802.11b	11 Mbps	2.4 GHz	Extension of original wireless standard
802.11g	54 Mbps	2.4 GHz	OFDM waveform. Backwards compatible with 802.11b
802.11n	300-600 Mbps	2.4 or 5 GHz	Introduces MIMO Up to four MIMO streams
802.11ac	Up to 7 Gbps	5 GHz	Introduces MU-MIMO and standardised beam-forming Up to eight MIMO streams
802.11ax	10 Gbps	All band spectrums between 1 and 7 GHz	MU-MIMO available for uplink as well as downlink Spatial frequency colouring and dynamic fragmentation

Drawbacks of Wi-Fi networking

Signal degradation: With distance, obstacles like walls can weaken Wi-Fi signals and connections.

Radiofrequency interference: Wi-Fi signals can clash with other devices on the same frequencies, such as cordless phones, microwaves, or Bluetooth.

Security vulnerabilities: Without strong encryption or authentication, Wi-Fi networks are at risk of unauthorised access and data interception.

Reduced performance: Crowded areas with many Wi-Fi networks can lead to channel interference, impacting speed and reliability.

Bandwidth sharing: Connecting multiple devices reduces the available bandwidth for each, potentially leading to slower internet speeds.

Wi-Fi frequencies

2.4 GHz

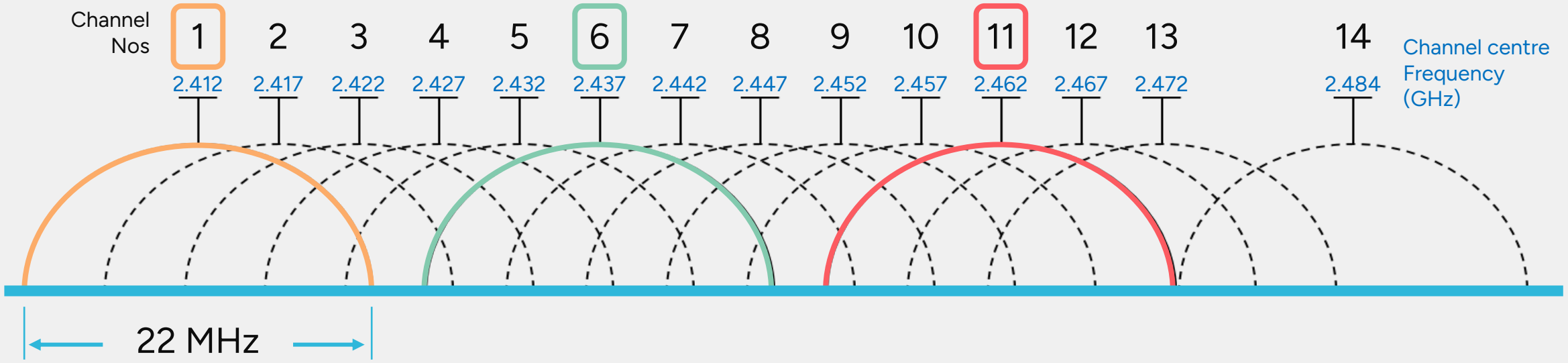
- **Advantages:** Offers a longer range and better ability to penetrate walls and obstacles.
- **Disadvantages:** More susceptible to interference from other devices and provides slower speeds.

5 GHz

- **Advantages**: Delivers faster speeds and experiences less interference due to a greater number of available channels.
- **Disadvantages**: Has a shorter range and less effective penetration through walls.

Wireless networking

2.4GHz 802.11b/g/n
What is the significance of channels 1, 6, and 11?



Wi-Fi security methods

WEP (Wired Equivalent Privacy) An older protocol with basic encryption.

- Weakness: Vulnerable to hacking due to weak encryption

WPA (Wi-Fi Protected Access) Improved security over WEP with dynamic encryption.

- Weakness: Outdated and vulnerable to modern attacks

WPA2 (Wi-Fi Protected Access 2) Stronger encryption based on AES.

- Weakness: Still vulnerable to some exploits like KRACK, but more secure than WEP or WPA

WPA3 (Wi-Fi Protected Access 3) Latest standard with enhanced encryption and authentication. Protection against brute-force attacks, better public Wi-Fi security, and forward secrecy.

- Weakness: Requires compatible hardware

Wireless networking

WPA2-Enterprise is a Wi-Fi security protocol designed for business and enterprise environments, providing a higher level of security compared to WPA2-Personal.

It uses 802.1X authentication with a RADIUS server (Remote Authentication Dial-In User Service) to authenticate each device or user before granting access to the network.

- **Strengths**: Strong security with unique user credentials
- **Weakness**: More complex setup and infrastructure

Wireless networking

An **SSID (Service Set Identifier)** is the unique name assigned to a Wi-Fi network, helping devices identify and connect to the correct network among many nearby.

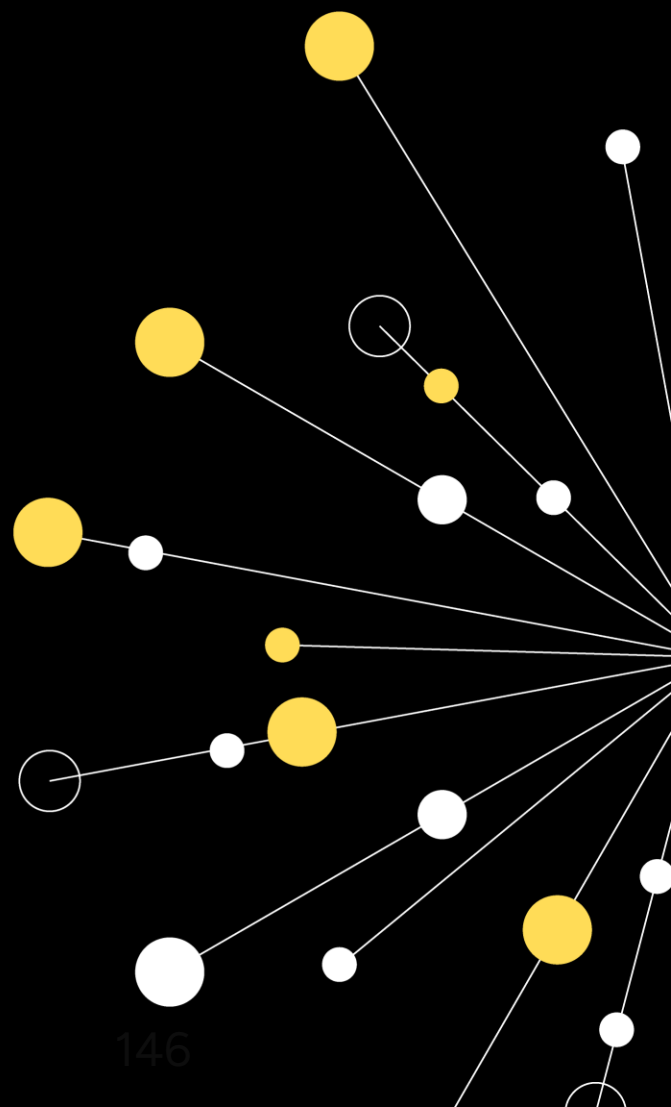
It serves as the identifier for a wireless local area network (WLAN).

- **Network identification:** Devices use the SSID to list available networks and connect to the right one.
- **Security:** SSIDs can be hidden, requiring manual entry to connect, though this doesn't significantly improve security.
- **Broadcasting:** By default, routers broadcast the SSID, making the network visible to nearby devices.
- **Length:** An SSID can be up to 32 characters long and may include letters, numbers, and special characters.

Activity 14: Wireless practical

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



In summary: Wireless networking

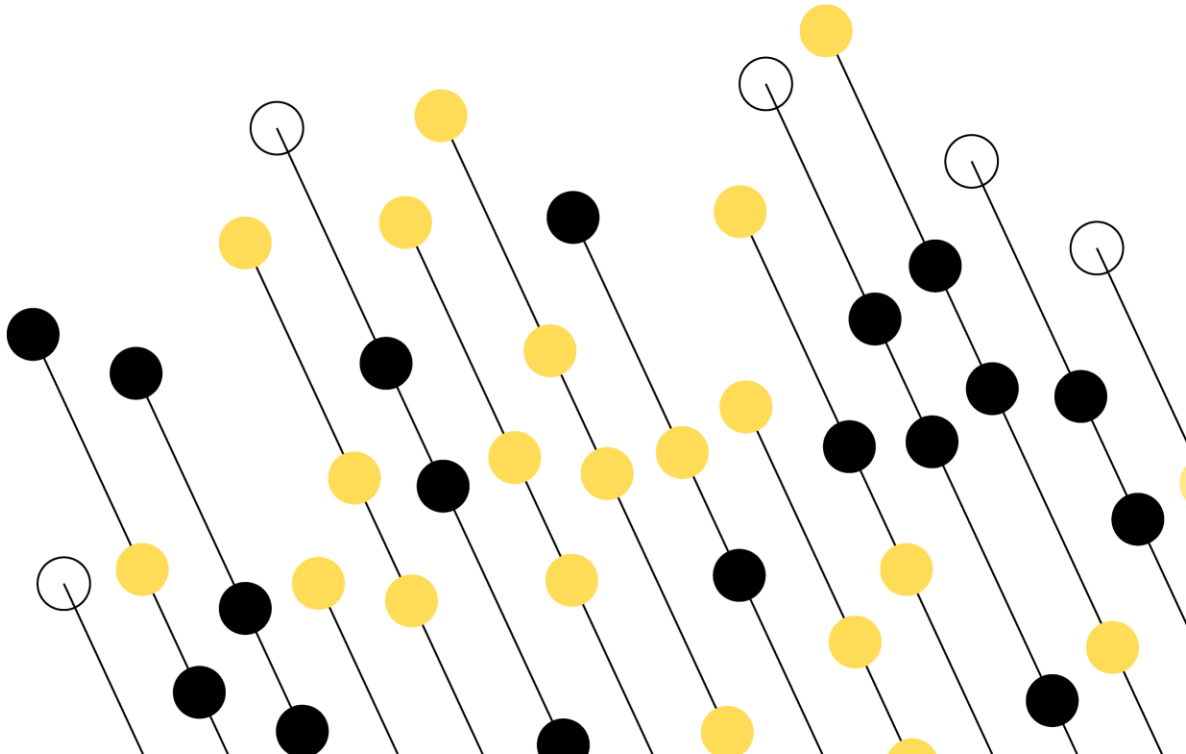
Wireless networking enables devices to connect and communicate without physical cables, offering convenience, mobility, and flexibility.

It operates on different frequency bands like 2.4 GHz and 5 GHz, each with its strengths and limitations.

Wi-Fi security methods, such as WPA2 and WPA3, ensure that wireless networks remain secure from unauthorised access and threats.

Understanding wireless technologies, security protocols, and their potential drawbacks is essential for maintaining reliable and secure wireless connections.

Internet Protocol version 4 (IPv4)



Internet Protocol version 4 (IPv4)

IPv4 uses 32 bits. The 32 bits are organised into four octets, with each octet consisting of eight bits: 8 (bits) x 4 (octets) = 32.

Example:

1st Octet	2nd Octet	3rd Octet	4th Octet
10101010	01010101	11111111	00000000

This structure allows for a range of IP addresses suitable for the needs of early internet growth.

IPv4: Address range

An IPv4 address consists of four octets, separated by dots (e.g., 192.168.1.10).

Each octet in an IPv4 address is can represent 256 unique values, ranging from 0 to 255 (8 bits can form 2^8 = 256 combinations).

The total number of possible IPv4 addresses can be calculated as:

$$256 \times 256 \times 256 \times 256 = 4{,}294{,}967{,}296$$

This means IPv4 can theoretically support approximately **4.3 billion unique addresses**.

IPv4: Address range

The lowest possible address is 0.0.0.0, and the highest is 255.255.255.255.

The lowest address, **0.0.0.0**, is represented in binary as:

00000000.00000000.00000000.00000000

The highest address, **255.255.255.255**, is represented in binary as:

11111111.11111111.11111111.11111111

IPv4: Address classes

The Internet Assigned Numbers Authority (IANA) ensures that the assignment of IP addresses is done in a structured and orderly manner.

This avoid conflicts and ensure proper routing across the internet.

IPv4 addresses are divided into five classes (A, B, C, D, and E), each designed for different purposes and types of networks.

These classes help organise the distribution of IP addresses efficiently based on the requirements of various types of networks and devices.

IPv4: Class A (1.0.0.0 to 127.255.255.255)

Class A is reserved for very large networks, such as those used by large corporations or government organisations.

- **First octet:** Starts with a bit pattern of 0 (binary), meaning that the first octet ranges from 1 to 127.
- **Network/host division:** Class A allows for 128 networks, each with over 16 million hosts.

Class B (128.0.0.0 to 191.255.255.255)

Class B is designed for medium-sized networks like universities or regional networks.

- **First octet:** Starts with a bit pattern of 10 (binary), so the first octet ranges from 128 to 191.
- **Network/host division:** Class B allows for 16,384 networks, each supporting up to 65,534 hosts.

Class C (192.0.0.0 to 223.255.255.255)

Class C is primarily used for smaller networks, such as those found in small businesses and homes.

- **First octet:** Starts with a bit pattern of 110 (binary), meaning the first octet ranges from 192 to 223.
- **Network/host division:** Class C allows for 2 million networks, each supporting up to 254 hosts.

Class D (224.0.0.0 to 239.255.255.255)

Class D is used for multicast addressing, which is a method of transmitting data to multiple recipients at once.

- **First octet:** Starts with a bit pattern of 1110 (binary), so the first octet ranges from 224 to 239.
- **Multicast:** Class D addresses are not used for standard network-to-host communication but for one-to-many communications.

Class E (240.0.0.0 to 255.255.255.255)

Class E is reserved for experimental purposes and is typically not used for general networking.

- **First octet:** Starts with a bit pattern of 1111 (binary), meaning the first octet ranges from 240 to 255.
- **Experimental:** Class E is reserved for future use or experimentation and is not typically allocated for public use.

IPv4

Class	Starting IP	End IP	128	64	32	16	8	4	2	1
A	0	127	0	0	0	0	0	0	0	0
B	128	191	1	0	0	0	0	0	0	0
C	192	223	1	1	0	0	0	0	0	0
D	224	239	1	1	1	0	0	0	0	0
E	240	255	1	1	1	1	0	0	0	0

IPv4

Class	Starting IP	End IP	128	64	32	16	8	4	2	1
A	0	127	0	1	1	1	1	1	1	1
B	128	191	1	0	1	1	1	1	1	1
C	192	223	1	1	0	1	1	1	1	1
D	224	239	1	1	1	0	1	1	1	1
E	240	255	1	1	1	1	1	1	1	1

IPv4: Subnet masks

A **subnet mask** is a 32-bit number used in computer networks to divide an IP address into two parts:

1. The **network portion**
2. The **host portion**

It is primarily used to determine which part of an IP address represents the network and which part can be used for devices (hosts) within that network.

IPv4: Subnet masks

The subnet mask 255.255.255.0 means that the first three octets (192.168.1) define the **network**, and the last octet (10) identifies the specific device or **host.**

Class C IP	192	168	1	10
Default Subnet mask	255	255	255	0
Subnet in Binary	11111111	11111111	11111111	00000000

IPv4

The subnet mask determines how much of the IP address is used for the network and how much is used for the devices (hosts) in that network.

See the next slide for a breakdown.

IPv4

Class A IP	10	5	4	1
Default Subnet mask	**255**	0	0	0
	Network	Host	Host	Host

Class B IP	172	16	1	10
Default Subnet mask	**255**	**255**	0	0
	Network	**Network**	Host	Host

Class C IP	192	168	1	10
Default Subnet mask	**255**	**255**	**255**	0
	Network	**Network**	**Network**	Host

IPv4

To simplify it you can think of a street analogy. The **subnet mask 255s** defines the street you live on, and the **0** defines the door number you/your device reside at.

Class C IP	**192**	**168**	**1**	**10**
Default Subnet mask	255	255	255	0
	Network	Network	Network	Host
Analogy		Oxford Street		10

So, in this analogy, **192.168.1** represents the **street/network**. Changing any of these numbers in the IP address means you're changing the **street/network** you live on.

The .**10** in the IP address, defined in the subnet mask as **0** is the **Host/House number**. This is where your device resides. You can change this number, which would mean you're moving to a different **host/house** but still staying on the same **street/network**.

IPv4

If we go up to Class B the same analogy can be applied, but this time the **first 2 Octets define the Street/Network** defined by the subnet mask where is 255.255. The 3rd and 4th octet, defined by 0s, gives us the Host/House number.

Class B IP	**172**	**16**	**1**	**10**
Default Subnet mask	255	255	0	0
	Network	Network	Host	Host
Analogy	Bond Street		1	10

So, in this example we are saying **'I live at door number 1.10 on 172.16 (Bond Street)'**.

In technical terms, **Host 1.10 on the 172.16 Network**.

IPv4

The same can be applied for Class A, where the first Octet would define the Network/Street, and the remaining 3 Octet define the hosts/House.

Class A IP	**10**	**5**	**4**	**1**
Default Subnet mask	255	0	0	0
	Network	Host	Host	Host
Analogy	Regents Street	5	4	1

In this example we are stating we live at **5.4.1 Regents Street**, or in technical terms the **Host is 5.4.1 on the 10. Network**.

IPv4

So, what does it all mean knowing that information? Well, it comes down to the number of hosts you wish to support. This is where we need to understand the difference between **hosts and addresses**.

Looking at our Class C IP 192.168.1.10, we know what portions are a **host and network**. But we need to find out the address range. The **Lowest/First address** we can have from this network is **192.168.1.0**.

If we go lower than this, we would be changing the network all together, for example 192.168.0.255, which is not what we want.

The **Highest/Last address** we can have would be **192.168.1.255**.

So, our address range is going to be **192.168.1.0 – 192.168.1.255**.

This will give us 256 addresses in total.

IPv4

If you want a different viewpoint on this, use Excel or any spreadsheet software to help you:

1. Open Excel and go to Cell A:1 and type the first IP address: 192.168.1.0
2. Then, move your mouse to the bottom corner of the cell, where it will change to a crosshatch **+**
3. **Click, hold, and drag** down 256 cell values.
4. You will now see the last address being 192.168.1.255 in cell A:256.

IPv4

Now that we have determined we have 256 addresses, the next part is to work out the number of **hosts**.

All hosts are addresses... but not all addresses are hosts!

This is where we introduce the **Network ID** and the **Broadcast**.

The **Network ID** will be used at the **.0 location** of the IP and the **Broadcast** will use the **.255 location**.

This means we will have 254 hosts and 256 addresses.

Address	Type
192.168.1.0	Network ID
192.168.1.1	First host
192.168.1.254	Last host
192.168.1.255	Broadcast

Network ID

Network ID defines the network's starting point and is used to route traffic to the correct subnet.

It represents the entire network and not any specific device.

It cannot be assigned to devices (like computers or printers).

Routers and network devices use it to direct packets to the correct subnet, ensuring efficient data flow.

Broadcast address

Broadcast address allows communication with all devices in the network simultaneously, typically used for announcements or discovery protocols.

It is reserved and cannot be assigned to a specific device.

It enables one-to-all communication within a subnet, making it ideal for broadcasting information efficiently.

This is used by network administrators for tasks like troubleshooting and network scans.

IPv4

Referring to our analogy, the **Network ID** can be compared to a **street sign** that identifies the street you live on.

Just like a street sign indicates the name of the street and helps people find their way, the Network ID identifies the specific network within the broader system of interconnected networks.

In our case the Network ID is set at location .0

IPv4

The **broadcast address** can be compared to a **village noticeboard** you might find in a local park.

Just as a noticeboard is a centralised place to post messages that everyone in the village can see, the broadcast address is a special IP address that allows messages to be sent to all devices within a network.

In our example it is in location .255

IPv4

A **house** can represent a **host**. Each house is identified by its door number.

Just like many houses can exist on a single street, you can have multiple hosts on the same network.

- 192.168.1.1 = House number 4
- 192.168.1.2 = House number 5

IPv4

As mentioned, the Internet Assigned Numbers Authority (IANA) is the global organisation responsible for managing and allocating IP addresses.

To ensure consistency and proper functioning of networks worldwide, IANA has established specific guidelines and address ranges that need to be followed.

These ranges are categorised into different types of addresses, including **Private IP addresses**, **APIPA (Automatic Private IP Addressing)**, and **Loopback addresses**.

IPv4: Private IP addresses

Private IP addresses are reserved for use within private networks, such as homes, offices, or organisations.

These addresses are not routable on the public internet, meaning they can only communicate within their local network.

IANA has defined the following private IP ranges:

- **Class A: 10.0.0.0 to 10.255.255.255**
- **Class B: 172.16.0.0 to 172.31.255.255**
- **Class C: 192.168.0.0 to 192.168.255.255**

These ranges allow devices on private networks to communicate without conflicting with public IP addresses.

IPv4: APIPA

APIPA is a feature that allows devices to self-assign an IP address when they are unable to obtain one from a DHCP server.

This range is defined as:

- **169.254.0.0** to **169.254.255.255**

If a device cannot connect to a DHCP server (e.g., due to network issues), it automatically assigns itself an address within this range.

This ensures the device can still communicate with other devices on the same network segment, but it cannot communicate outside that local network.

APIPA is often used as a fallback mechanism for small or isolated networks when DHCP services are unavailable.

IPv4: Loopback address

The **loopback address** is a special address range reserved for internal testing and diagnostics on a device. This range is defined as:

- **127.0.0.0** to **127.255.255.255**

Most commonly, **127.0.0.1** is used to represent the local machine (localhost).

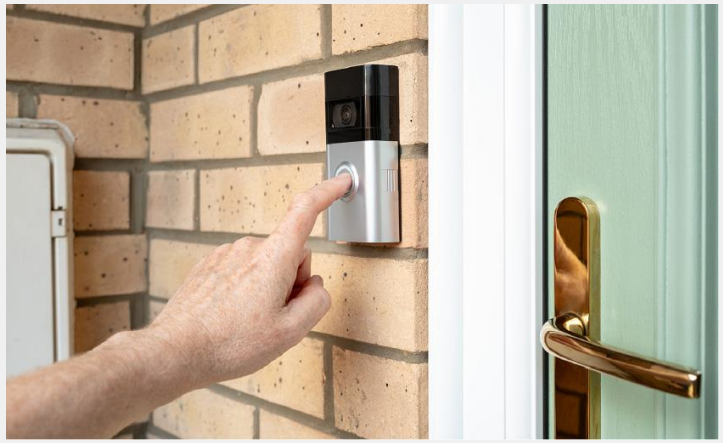
When a device sends data to the loopback address, the data is sent to itself. This allows users and applications to test network functionality without the need for external connections.

Loopback addresses are often used to troubleshoot network issues, test applications, or verify that TCP/IP is configured properly on a system.

IPv4

In terms of the simple analogy:

Private addresses can be seen as a gated community strictly for the owner.



Loopback could be seen as pressing your own home doorbell.

APIPA can be viewed as being in the forest where there is minimal resources.

Activity 14: IPv4 practical

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.

In summary: IPv4

IPv4 is the foundation of modern networking, providing a structured way to assign unique addresses to devices and facilitate communication across networks.

With its 32-bit addressing system, IPv4 supports approximately 4.3 billion unique addresses, divided into network and host portions.

Key concepts include **subnetting** for efficient IP management, **Network IDs** and **Broadcast Addresses** for organising and routing traffic, and specialised ranges like **Private IPs**, **APIPA**, and **Loopback** addresses for specific use cases.

Despite its limitations, such as address exhaustion, IPv4 remains widely used due to its simplicity, reliability, and compatibility with countless devices and systems globally.

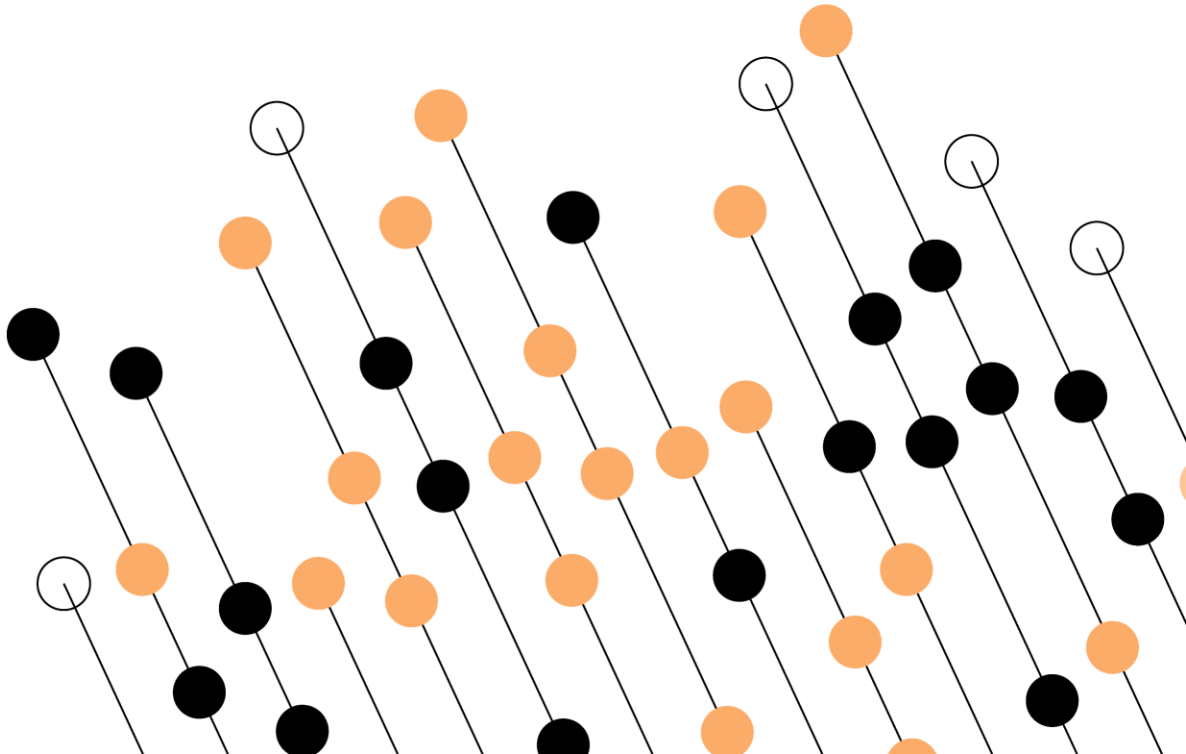
Core Cloud and Networking Skills

Day 4

QA

Learn. To Change.

Internet Protocol version 6 (IPv6)



IPv6

IPv6 is the modern internet protocol created to succeed IPv4 and address its shortcomings.

It has a larger address space, enhanced security, and other advanced capabilities.

IPv6 uses a 128-bit addressing system, allowing for 2^{128} unique addresses.

This equates to approximately **340 undecillion addresses**.

IPv6

IPv6 addresses are made up of **eight hextets** separated by **colons**.

Each hextet has **four hexadecimal digits**, and each **digit uses four bits:**

IPv6 address:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Converted into binary:
0010 0000 0000 0001 : 0000 1101 1011 1000 : 1000 0101 1010 0011 : 0000 0000 0000 0000 : 0000 0000 0000 0000 : 1000 1010 0010 1110 : 0000 0011 0111 0000 : 0111 0011 0011 0100

IPv6 features

Built-in security: Includes **IPsec (Internet Protocol Security)** as a mandatory feature for authentication and encryption of data packets.

Auto-configuration: Supports stateless address autoconfiguration (SLAAC), enabling devices to automatically generate their own IP addresses without the need for a DHCP server.

Removes broadcast: Better multicast communication (sending a single packet to multiple destinations) and adds anycast addressing, where packets are sent to the nearest device in a group.

IPv6: Address types

IPv6 addresses are categorised into several types based on their functionality and purpose.

Unicast addresses are used to identify a single device on a network. Traffic sent to a unicast address is delivered to the specific interface assigned that address.

There are several subtypes of unicast addresses.

IPv6: Unicast addresses

- **Global Unicast Address (GUA):** Routable on the global internet. Begins with prefixes such as 2000::/3 **Example**: 2001:0db8::1

- **Unique Local Address (ULA):** Used for communication within a private network and are not routable on the internet. Starts with FC00::/7 **Example**: FD00::1

- **Link-Local Address:** Automatically configured on IPv6-enabled interfaces for communication within a single link or network segment. Starts with FE80::/10. **Example**: FE80::1

IPv6: Multicast and anycast addresses

Multicast addresses deliver traffic to multiple devices simultaneously, typically for group communication.

A single packet is sent, and all devices subscribed to the multicast group receive it.

This makes them efficient for applications like video conferencing or streaming.

They always start with FF00::/8

- **Example**: FF02::1 (all nodes on the local link).

Anycast addresses are like multicast but are only delivered to the closest member of the group (based on routing metrics).

It is often used for load balancing and redundancy.

It assigns the same address to DNS servers globally, and clients are routed to the nearest server.

IPv6: Reserved addresses

- **Loopback address:** Used for testing on the local device.
  - **Address:** ::1

- **Unspecified address**: Represents an absence of an address (used in initial stages of communication before an address is assigned).
  - **Address:** ::

- **IPv4-Mapped IPv6 address:** Allows IPv6 devices to communicate with IPv4 devices by embedding an IPv4 address in the IPv6 address. ::FFFF:0:0/96
  - **Example:** ::FFFF:192.168.1.1

IPv6: Address types summary

First hextet	Type of IPv6 address
0000 to 00FF	Loopback address, any address, unspecified address, or IPv4 – compatible
2000 to 3FFF	Global unicast address (a routable address in a range of addresses issued by the Internet Assigned Numbers Authority (IANA))
FE80 to FEBF	Link-local (a unicast address which identifies the host computer on the local network)
FC00 to FCFF	Unique-local (a unicast address which can be assigned to a host to identify it as part of a specific subnet on the local network)
FF00 to FFFF	Multicast address

IPv6

Here is an expanded explanation of the breakdown of a global unicast IPv6 address into its three main parts: **Global routing prefix**, **Subnet**, and **Interface ID**.

The global routing prefix is the first portion of the IPv6 address. It identifies the network to which the address belongs. This section is used by routers to forward packets to the correct destination network on the global internet.

Typically, this is the first 48 bits (or more) of the IPv6 address. This prefix is assigned by an internet service provider (ISP) or a central authority.

Analogy: Think of this as the ‘city’ identifier in a mailing address – it directs the traffic to the correct geographic location.

IPv6

The IPv6 subnet is the middle portion of the address, used to define smaller subdivisions (subnets) within the larger network defined by the global routing prefix.

This allows network administrators to organise and manage their networks efficiently, segmenting traffic for performance or security purposes.

Often the next 16 bits (after the global routing prefix) are used for subnetting, but this can vary depending on network needs.

Analogy: Consider this as the ‘neighbourhood’ in a mailing address – it specifies a more specific area within the city.

IPv6

The interface ID is the final portion of the IPv6 address, identifying a specific device or interface within a subnet.

This is the unique identifier for a device on the network, ensuring that each device can communicate without conflict.

Usually, this is the last 64 bits of the IPv6 address. These can be manually assigned, dynamically assigned, or generated using a method like EUI-64.

Analogy: Think of this as the ‘house number’ in a mailing address – it pinpoints the exact location within the neighbourhood.

IPv6

2001:0db8:1234:abcd:0000:0000:0000:0001

Global Routing Prefix: 2001:0db8 (identifies the larger network)

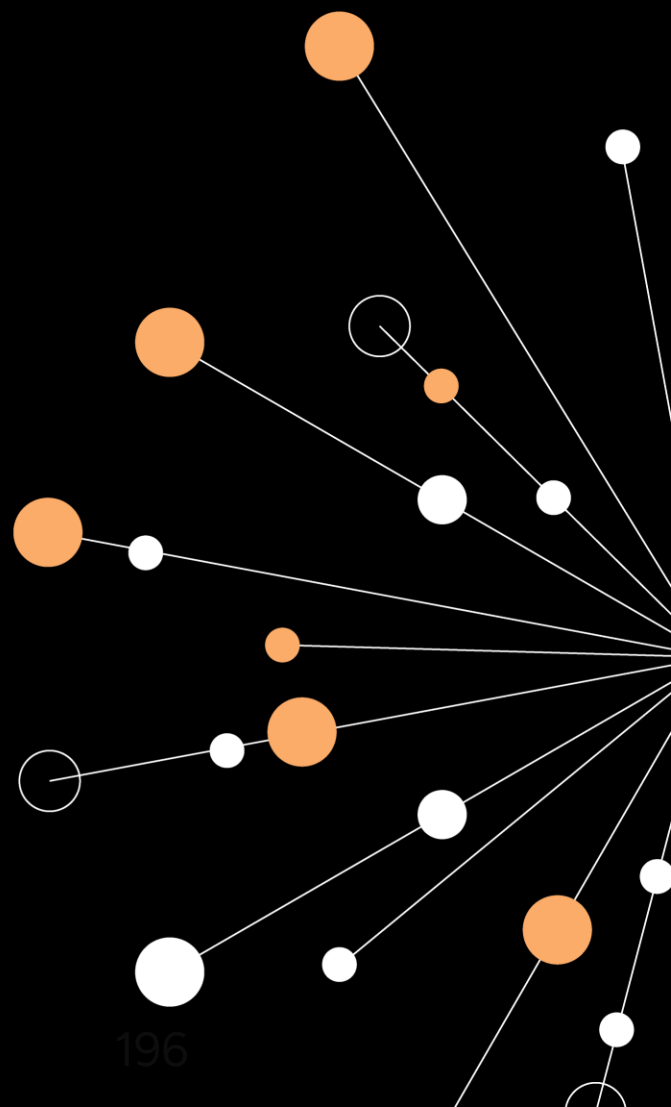
IPv6 Subnet: 1234:abcd (defines the specific subnet within that network)

Interface ID: 0000:0000:0000:0001 (uniquely identifies the device within the subnet)

Activity 15: IPv6 Q&A

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



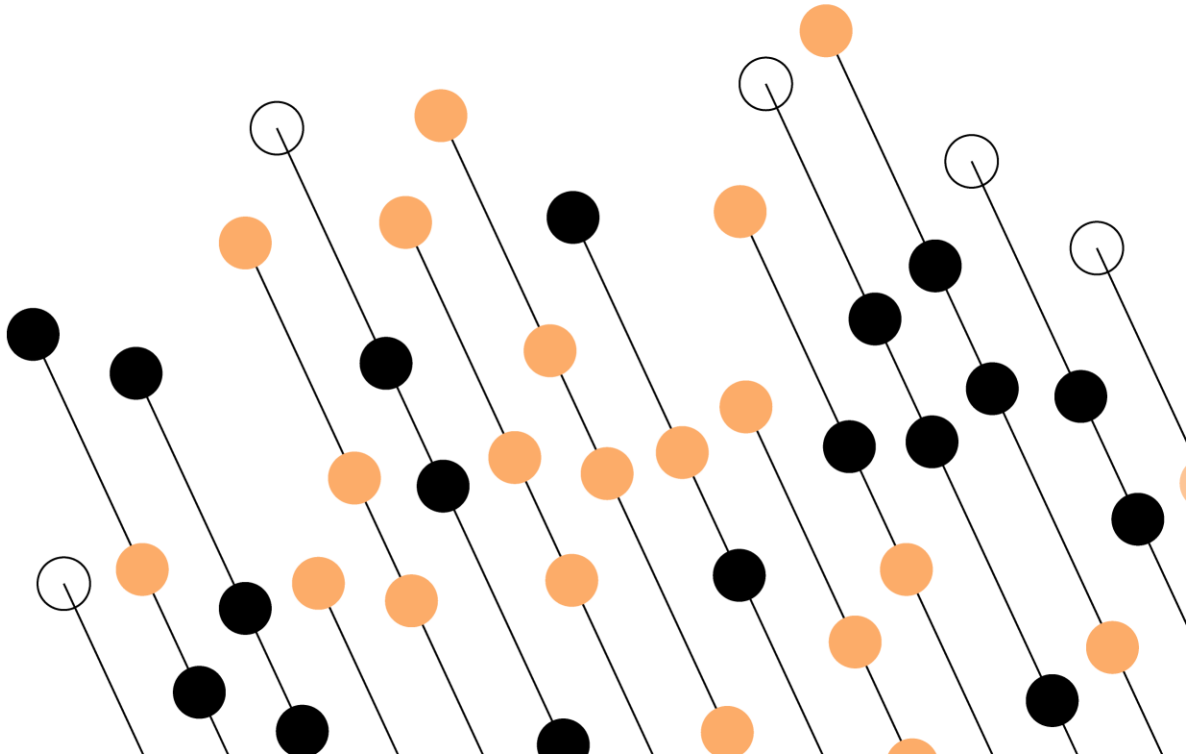
In summary

IPv6 is the next-generation internet protocol designed to address the limitations of IPv4, offering a virtually limitless address space, enhanced security, and improved efficiency.

With features like simplified address configuration, better multicast support, and the ability to handle modern internet demands, IPv6 ensures scalability and reliability for the growing digital world.

As its adoption continues, IPv6 plays a critical role in enabling the seamless operation of the internet for years to come.

Network services



Network services

Network services like DNS, DHCP, and the Default Gateway are fundamental components of modern networking infrastructures.

These services work together to ensure devices on a network can effectively communicate with one another, access resources, and reach external networks, such as the internet.

Each service plays a specific and crucial role in the overall functioning of a network.

Network services

The **Domain Name System (DNS)** is often referred to as the 'phonebook' of the internet, because it is responsible for translating human-friendly domain names into machine-readable IP addresses. Without DNS, users would have to remember a long string of numbers (IP addresses) for each website they want to visit, instead of easily remembering domain names like www.example.com.

The DNS server holds a large database of mappings between domain names and IP addresses. If the server does not have the requested IP address cached, it will send a query to other DNS servers until the address is found.

Once the IP address is resolved, the device can initiate communication with the server hosting the website.

Network services

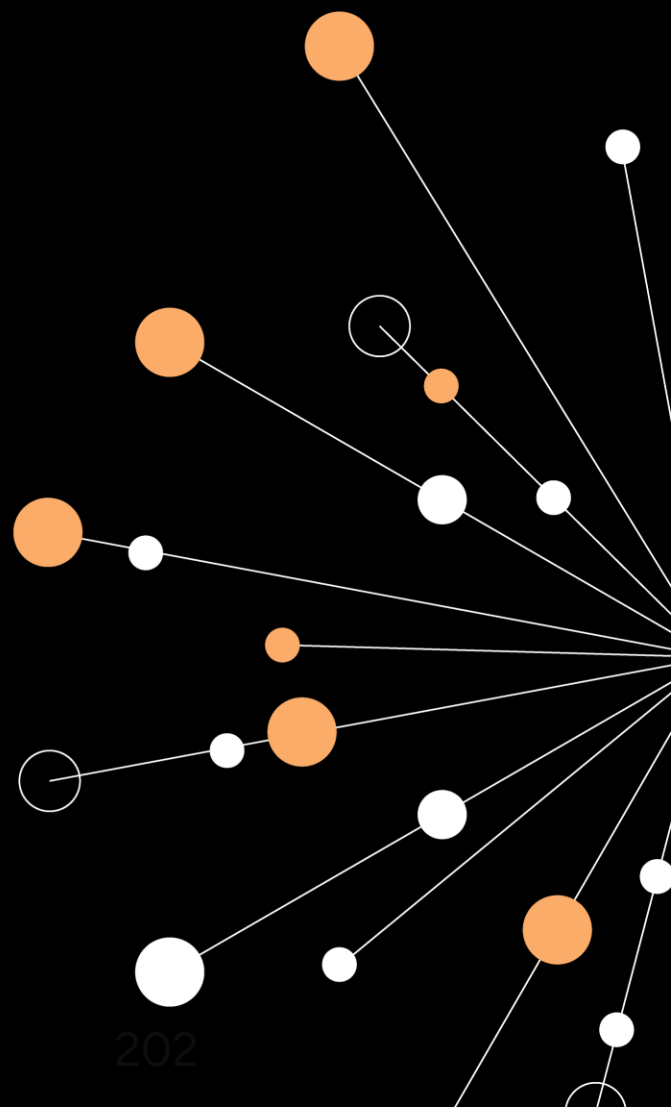
Public DNS is ideal for users who want a fast, reliable, and secure DNS service for general internet browsing. It’s commonly used by home users, businesses, and anyone needing to access websites on the public internet.

Private DNS is best suited for organisations, businesses, or home networks that need control over internal domain name resolution, security, and traffic routing within a private environment. It is used to manage local resources and restrict access to specific internal services while providing enhanced security and privacy features for users within the network.

Activity 16: DNS practical

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Network services

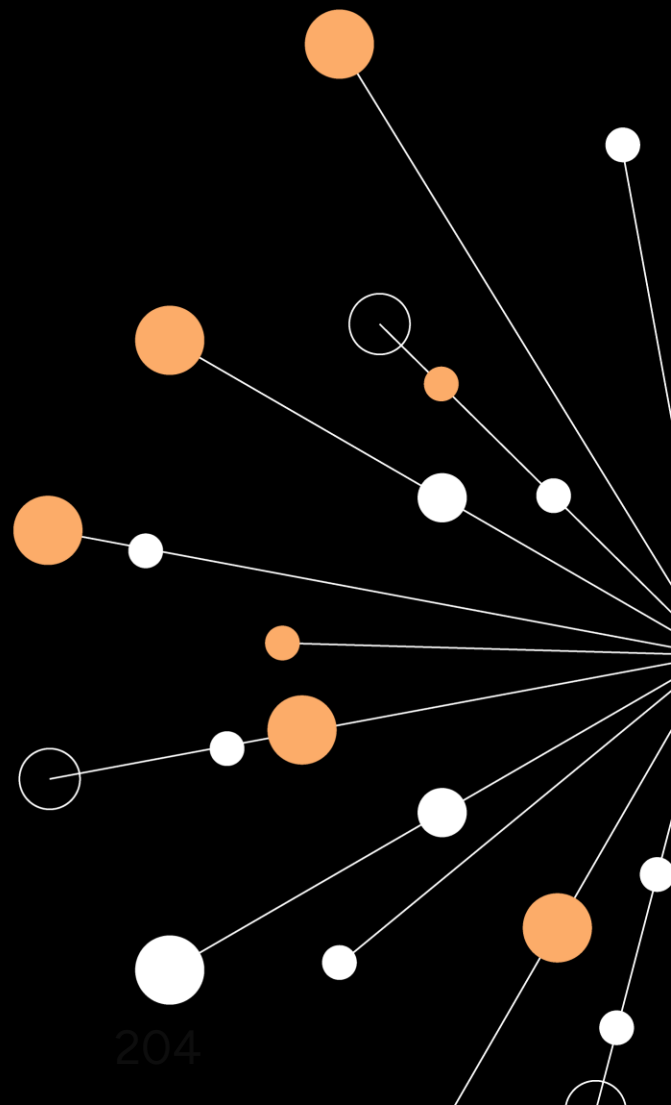
DHCP (Dynamic Host Configuration Protocol) automates the process of assigning IP addresses to devices on a network. In modern networks, especially large ones, manually assigning IP addresses to every device would be time-consuming and error-prone. DHCP simplifies this by allowing devices to request an IP address from a central server.

The DHCP server assigns an IP address, a subnet mask, a default gateway, and DNS servers to the device for a specified lease period. Once the lease expires, the device must request a new address.

Activity 17: DHCP practical

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Network services

The **Default Gateway** is the network device (often a router) that forwards traffic from a local network to other networks, typically the internet.

Without a default gateway, devices within a local network would only be able to communicate with each other.

The gateway enables them to reach external resources by routing traffic to and from different networks.

For devices within a home or office network, the router often serves as the default gateway and is responsible for forwarding internet-bound traffic between local devices and the internet.

In summary: Network services

Network services like DNS, DHCP, and the default gateway are essential for enabling effective communication and resource access within and beyond a network.

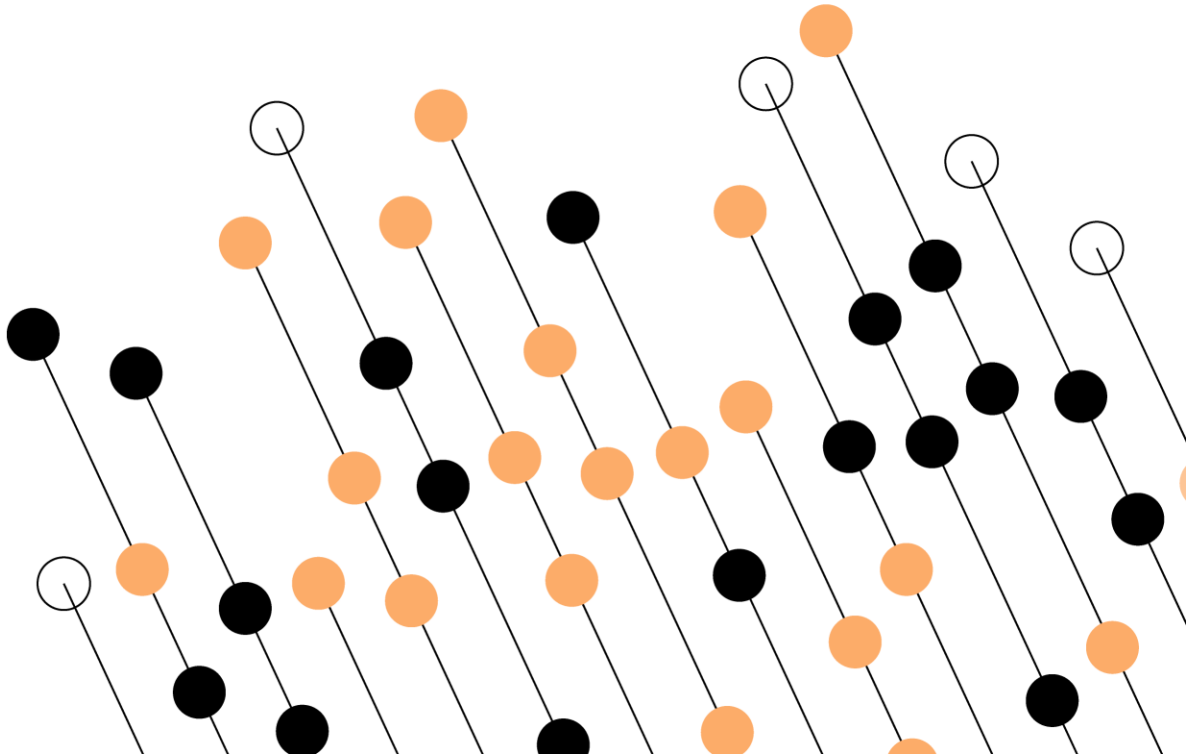
DNS translates domain names into IP addresses, simplifying access to websites and services.

DHCP automates the assignment of IP addresses to devices on a network, reducing the need for manual configuration.

The default gateway routes traffic from the local network to external networks, such as the internet.

Together, these services ensure seamless and secure connectivity, allowing devices to communicate efficiently within a network and with the broader internet.

Command Prompt



Command Prompt

The Command Prompt (often referred to as CMD) is a **command-line interpreter** built into Windows operating systems.

It allows IT professionals to execute text-based commands to perform various administrative, diagnostic, and troubleshooting tasks.

Within an IT department, the Command Prompt is a powerful tool due to its flexibility, speed, and ability to automate complex operations.

Command Prompt

Syntax refers to the structured format or rules that a command must follow to be understood and executed by the Command Prompt.

command [options] [arguments]

- **Command:** The primary instruction or action to be performed
- **Options (or switches):** Modify or extend the behaviour of the command
- **Arguments:** Provide additional information needed for the command, such as file names, directories, or parameters

ping -n 4 www.google.com

- **ping:** The command to test network connectivity
- **-n 4:** An option specifying that 4 packets should be sent
- **www.google.com:** The argument, indicating the target domain to ping

Command Prompt example

ping -n 4 www.google.com

- **Command**: ‘ping’ (the command to test network connectivity)
- **Option:** ‘-n 4’ (an option specifying that 4 packets should be sent)
- **Argument:** ‘www.google.com’ (the argument, indicating the target domain to ping)

Command Prompt: The change directory command

- **cd c:\users\administrator\desktop** changes the prompt to the stated directory.
- **cd..** goes to the parent folder.
- **cd/?** describes the command and show you what options are available.

Command Prompt: The dir command

Use **dir** to list directory contents:

Options	Use
dir /p	Displays one screen of the listing at a time – to see the next screen, press any key on the keyboard Best to be used in folder with a lot of content like \system32
dir /q	Displays file ownership information
dir /q	Displays the listing in wide format, with as many as five file names or directory names on each line

Command Prompt: The dir command

dir /s allows searching for files or folder names.

You can use wild cards as part of the names:

- ***** represents any number of characters.
- **?** represents a single character.

Try this as an example:

- **dir /s *.png** (will display all .png files)
- **dir /s ????.png** (will display all .png files with the file name being four characters long)

Command Prompt: The dir command

Sort contents with **dir /o**:

Option	Purpose (sort by...)
n	Name (alphabetical)
e	Extension (alphabetical)
g	Group directories first
s	Size (smallest first)
d	Date/time (older first)
-	Prefix to reverse order

- **dir /o:n**
- **dir /o:-n**

Command Prompt: The mkdir command

This will allow you to create a folder in the current location, md also works in the same way. This does not have switches or options.

Try the below as example:

- **mkdir test** (if the folder name is one word no quotes required)
- **mkdir “test one”** (if the folder name has two words quotes are required)

Command Prompt: The rmdir or rd command

These are used for deleting folders. Two possible switches:

Options	Use
/s	Removes the specified directory and all subdirectories including any files
/q	Runs in quiet mode and deletes directories without confirmation

Try this as an example:

- **rmdir c:\user\guest\desktop\test** (if the folder name is one word no quotes required)
- **rmdir c:\user\guest\desktop\"test one"** (if the folder name has two words quotes are required)

Command Prompt: The del command

The del command is used for erasing files. Possible switches:

Switch	Use
/p	Prompts for confirmation before deleting the specified file
/f	Forces deletion of read-only files
/s	Deletes specified files from the current directory and all subdirectories Displays the names of the files as they are being deleted
/q	Specifies quiet mode You are not prompted for delete confirmation

Try this as an example:

1. **type nul > test.txt** (will create a file)
2. **del /p test.txt** (will delete the file)

Command Prompt: chkdsk, sfc, and dism commands

The chkdsk command

Windows uses chkdsk to check the disk for errors:

- **chkdsk C: /f** (**/f** option automatically fix errors)

System File Checker (SFC) tool

It runs a scan for corrupted or missing system files and attempts repairs using cached copies:

- **sfc /scannow**

Deployment Image Servicing and Management (DISM) tool

Used for servicing and repairing Windows images:

- **dism /online /cleanup-image /scanhealth** (checks for component store corruption in the Windows image)
- **dism /online /cleanup-image /checkhealth** (quick assessment without initiating repairs – if corruption is detected, you can proceed with other DISM commands for repairs)

Command Prompt: Useful commands summary

- **IPConfig:** Shows the IP configuration of the system you are working on
- **Ping:** Check if another node is active
- **TraceRT**: Identifies the router is which a packet takes to reach a destination
- **PathPing:** Pings a node IP address but also logs the router the packet takes
- **NetStat:** Shows the user the active connection on their network card
- **Telnet:** Unsecure remote shell

FTP: Allows a user to access an FTP server

NbtStat: Similar to Netstat, only for Netbios machines

Net Start / Net Stop: Start and stop the network services on a machine

NSLookUp: Query and troubleshoot name resolution

NetSh: Secure remote shell

Route: Display and configure the routing table

Command Prompt: The ipconfig command

Ipcolpconfig and ping can be used to analyse, test, troubleshoot, an configure IPv4 and IPv6 connections. **Ipconfig** options:

Option	Use
/all	Displays detailed network configuration information for all network interfaces on the system
/release	Releases the current IP address assigned to a network adapter by the DHCP server
/renew	Requests a new IP address from the DHCP server for the specified adapter or all adapters
/flushdns	Clears the DNS resolver cache, which stores recently resolved domain names
/registerdns	Refreshes all DHCP leases and re-registers the system's DNS name with the DNS server

Command Prompt: The ping command

Used to test network connectivity of your computer or host, verifying network connectivity between devices. Also used for testing DNS resolution by pinging hostnames, diagnosing slow or intermittent connections, and checking whether firewalls or filters are blocking ICMP packets.

Option	Use
-t	Sends continuous ping requests until stopped manually (Ctrl + C).
-n *	count Specifies the number of echo requests to send (**ping -n 5 google.com**).
-l size *	Sets the size of the packet in bytes (**ping -l 1000 google.com**).
-4	Forces the use IPv4
-6	Forces the use of IPv6
-a	Resolves an IP address to its hostname, if possible.

Command Prompt: The netstat command

The netstat command (short for 'network statistics') is a powerful utility used to display detailed information about network connections, routing tables, and various network interface statistics. It is commonly used for monitoring and troubleshooting network-related issues.

Options	Use
-a	Displays all active connections and listening ports
-n	Displays addresses and port numbers in numeric
-b	Shows the executable or program associated with each connection

Try this as an example as administrator – is Searchhost.exe safe?

netstat –anb

Command Prompt: The tracert and pathping commands

The tracert command is a diagnostic tool used to determine the path that data packets take to reach a specific destination. It shows each hop (router or device) along the way and measures the time it takes for packets to travel to each hop.

Try this as an example:

tracert google.com

The pathping command is an advanced network diagnostic tool that combines the functionality of **ping** and **tracert**. It provides a detailed report about the route to a destination, including packet loss and latency statistics for each hop.

Try this as an example:

pathping google.com

Command Prompt: The nslookup command

This is short for ‘Name Server Lookup’ and is a diagnostic tool used to query DNS servers and obtain domain name or IP address mapping. It helps verify that DNS servers are correctly resolving domain names and provides detailed information about DNS records.

Try the below as example:

Nslookup
Set type=a
google.com

Set type=aaaa
facebook.com

Command Prompt: The netsh command

This is a versatile command-line tool for managing and troubleshooting network configurations on Windows. Its broad range of commands for modifying network interfaces, DNS settings, firewalls, and routing tables makes it an essential tool for network administrators and IT professionals.

Try the below as example:

netsh interface show interface

netsh wlan show profiles

Command Prompt: The route print command

The route print command is an essential tool for diagnosing network routing issues and inspecting how network traffic is being handled on a system. By displaying the active routes, gateways, metrics, and interfaces, it provides valuable insights into the network configuration, which is helpful for network troubleshooting

Try the below as example:

route print

Command Prompt: The net command

The net command is a powerful and essential tool for network and system administrators to manage network resources, services, and user accounts from the command line.

It provides a broad range of functionalities, from managing shared resources to controlling services and mapping network drives, making it an asset in network configuration.

Add a new user: net user JohnDoe password123 /add

List all network shares: net share

Start the Print Spooler service: net start spooler

In summary: Command Prompt

The Command Prompt is a powerful tool in Windows that allows users to execute a variety of commands to manage, configure, and troubleshoot system settings.

It provides access to numerous utilities **like ipconfig, ping, netstat, and netsh**, among others, which can help with network configuration, file management, system diagnostics, and more.

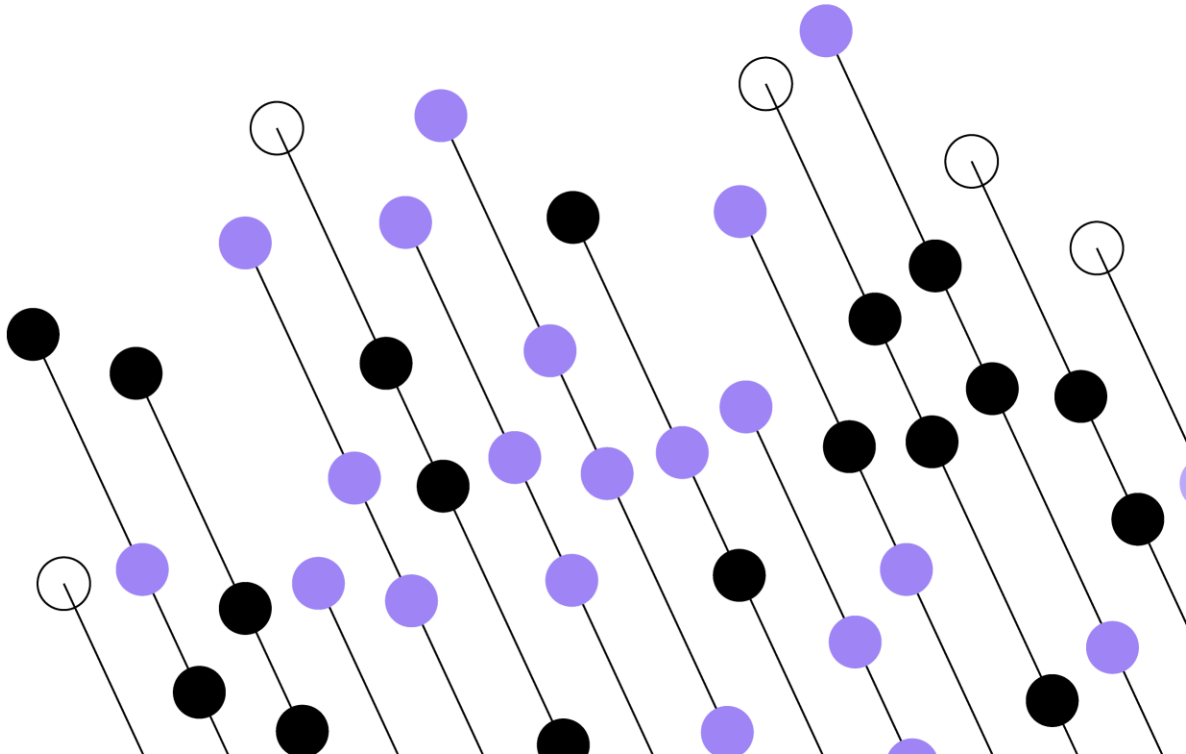
The Command Prompt is especially useful for IT professionals and system administrators as it allows quick, direct interaction with the operating system, offering more control and precision compared to graphical interfaces.

Core Cloud and Networking Skills

Day 5

Learn. To Change.

Troubleshooting



Troubleshooting: Problem identification

Define the issue clearly: Understand the symptoms of the problem, whether it's network connectivity issues, cloud service disruptions, or hardware failure.

Reproduce the problem: If possible, try to recreate the issue to understand its behaviour and context.

Gather details: Collect relevant data, including error messages, system logs, and network performance metrics.

Troubleshooting: Gathering information

Network configuration: Review IP configurations, DNS settings, routing tables, and firewall rules.

Cloud resources: Check the status of virtual machines, cloud storage, networking components, and cloud security configurations.

Logs and monitoring: Use network monitoring tools, cloud monitoring dashboards, and system logs to track events leading to the issue.

User feedback: Gather insights from end users to understand the scope and impact of the problem.

Troubleshooting: Diagnostic tools and techniques

Traceroute (tracert): Identify the path data takes across the network and locate where issues like latency or packet loss occur.

Netstat: View active network connections and determine whether the issue is related to port availability or open connections.

ipconfig: Check IP configuration on local devices to ensure correct addressing, subnetting, and gateway settings.

DNS resolution tools (nslookup, dig): Diagnose DNS issues to ensure that domain names resolve correctly to the appropriate IP addresses.

Cloud service dashboards and logs: Use tools like AWS CloudWatch, Azure Monitor, or Google Cloud Monitoring to identify performance issues, resource consumption, and error logs.

Ping: Check basic network connectivity between devices (e.g., between local machines and cloud servers).

Troubleshooting: Isolating the issue

Isolate the network layer: Check if the issue is in the physical network (hardware failure, cable issues) or at the software level (misconfigured DNS, IP conflicts).

Segmentation: Break down the system into smaller components (e.g., network layer, application layer, hardware) to pinpoint the exact source of failure.

Rule out external factors: Ensure the issue is not related to external internet service providers, upstream cloud providers, or third-party services.

Troubleshooting: Resolution and fix implementation

Configuration corrections: Modify network configurations, IP addresses, routing tables, DNS settings, or cloud resources to correct issues.

Restart services or devices: In cases where a service or device has become unresponsive, restart network devices, virtual machines, or cloud services.

Update software: Apply necessary patches or updates to systems, applications, or networking devices to resolve bugs or compatibility issues.

Rebuild cloud resources: In cloud environments, if a virtual machine or container is corrupted, consider redeploying or restoring from backups.

Troubleshooting: Verification

Testing: After applying fixes, conduct tests to verify that the issue is resolved. This can include checking network connectivity, application accessibility, or cloud resource performance.

Monitor systems: Continue to monitor systems for any recurrence of the issue and ensure everything is functioning as expected.

Troubleshooting: Documentation and reporting

Document the troubleshooting process: Record steps taken, tools used, and the resolution applied for future reference.

Report to stakeholders: Communicate findings and resolutions to relevant stakeholders, including users, management, or other IT teams.

Post-mortem analysis: Conduct a review of the incident to identify lessons learned and opportunities for improving system resilience or troubleshooting processes.

Troubleshooting: Preventive measures

Improve monitoring: Set up more robust monitoring and alerting systems to catch potential issues before they escalate.

Review network and cloud architecture: Reevaluate the design and configuration of the network or cloud resources to prevent recurring issues.

Implement automation: Use automation tools for configuration management and monitoring to reduce the likelihood of human error.

Troubleshooting: Root cause analysis

Analyse logs and metrics: Dive deep into logs and performance data to identify patterns and narrow down the root cause.

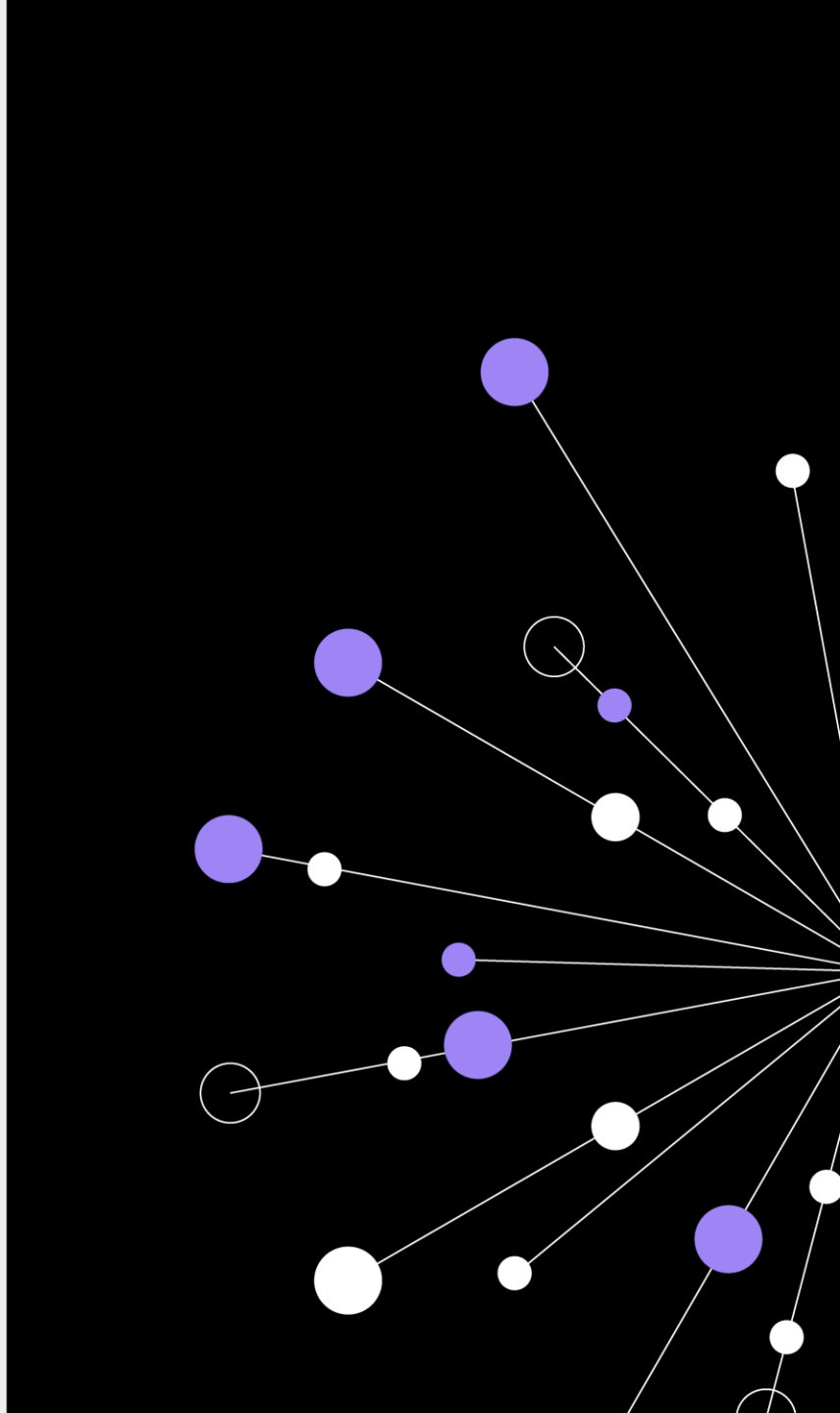
Check recent changes: Review any recent configuration changes, software updates, or network topology changes that may have contributed to the problem.

Cross-reference systems: Compare cloud and network system configurations with best practices and ensure compliance with architectural guidelines.

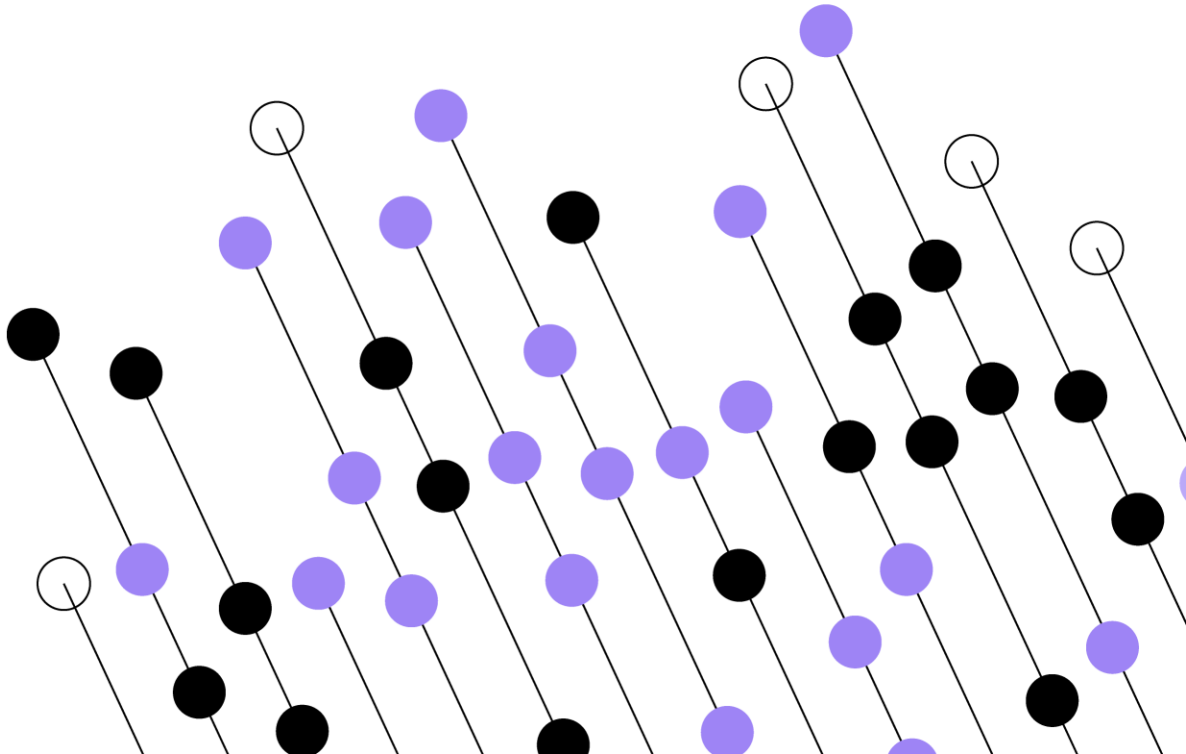
Activity 18: Troubleshooting practical

Complete the activity in your Exercise Guide. Then use Miro to share your answers and findings.

If you need any assistance just message the tutor.



Critical analysis



Critical analysis

Critical analysis involves a thoughtful evaluation of various systems, configurations, technologies, and methodologies used within cloud infrastructure and networking.

This process helps identify potential issues, risks, inefficiencies, and areas for improvement, ultimately guiding decisions that lead to better system performance, security, and scalability.

Critical analysis: Evaluation of cloud architecture

Cloud service models (IaaS, PaaS, SaaS): A critical analysis of the most appropriate cloud service model for a given use case is essential. For instance, IaaS (Infrastructure as a Service) offers more control but requires more management, while SaaS (Software as a Service) is easier to deploy but offers less customisation.

Cost-benefit analysis: Analyse the costs of running applications on cloud infrastructure versus traditional on-premises solutions. Consider operational costs, potential savings on hardware, scalability advantages, and resource efficiency.

Critical analysis: Network design and optimisation

Network topology: Critically evaluate the network topology, such as the arrangement of routers, switches, firewalls, and load balancers, to ensure efficient routing and minimal latency. Proper segmentation and the use of virtual LANs (VLANs) can help optimise performance and security.

Bandwidth and latency: Assess whether the current network infrastructure can handle required bandwidth and meet latency requirements. This is especially critical for cloud applications where high-speed data transfer is essential for performance.

Critical analysis: Cloud security analysis

Identity and access management (IAM): Assess IAM policies to ensure only authorised users and systems can access cloud resources. Critical analysis should focus on role-based access control (RBAC), least privilege principles, and multi-factor authentication (MFA).

Data encryption: Evaluate whether sensitive data, both at rest and in transit, is encrypted using the latest standards (e.g., AES-256). The effectiveness of cloud-native encryption solutions should also be critically examined.

Critical analysis: Network and cloud performance monitoring

Monitoring tools: Evaluate the tools used for monitoring network performance (e.g., SolarWinds, Nagios, or cloud-native tools like AWS CloudWatch and Azure Monitor). These tools should be capable of tracking performance metrics, such as throughput, response times, and error rates.

Service level agreements (SLAs): Analyse SLAs with cloud providers to ensure they align with the organisation's needs for uptime, support response time, and resolution. SLAs can provide valuable insights into the reliability and performance guarantees of cloud services.

Critical analysis: Disaster recovery and business continuity

Backup strategies: Evaluate the frequency and reliability of backups in both cloud and on-premises environments. Ensure that backups are encrypted and regularly tested for integrity.

Failover capabilities: Analyse cloud-based failover systems to ensure quick recovery in case of failure. This involves critically assessing redundancy across multiple data centres or cloud regions.

Critical analysis: Cloud cost management and optimisation

Resource utilisation: Critically analyse how cloud resources are being utilised. Over-provisioning can lead to unnecessary costs, while under-provisioning may affect performance. Tools like AWS Cost Explorer or Azure Cost Management can be used for optimisation.

Cost predictability: Evaluate cloud pricing models (pay-as-you-go vs. reserved instances) and predict long-term costs based on usage patterns. Proper budgeting and forecasting are critical in cloud environments.

Critical analysis: Documentation and best practices

Configuration management: Review the use of configuration management tools (e.g., Ansible, Terraform, or AWS CloudFormation) to automate cloud infrastructure deployment and configuration. Ensuring adherence to best practices in code and configuration is vital.

Change management: Critically analyse the change management processes to ensure all modifications to cloud and network environments are properly documented, tested, and implemented in a controlled manner.

End of module review

End of module review

- Approaches to problem solving
- Task tracking systems
- Maintenance approaches
- Security concepts
- Malware
- Encryption
- What is the cloud?
- Virtualisation
- Backup and storage
- Wireless networking
- Internet Protocol v4
- Internet Protocol v6
- Network services
- Command Prompt
- Troubleshooting
- Critical analysis

End of module review

Kindly upload your completed Exercise Guide and *Certification of Completion to BUD under the following module:

- Activity (Live Event)

*Certification of Completion can be obtained by going to the live event module in BUD and press '**start now**'. Then tick off the modules accordingly.

Learning instructions & materials

To complete this activity:

1. Attend a two-day live event (*), led by your tutor | Start now | - **please don't start this learning path before the live event.**
2. Submit your work in the 'Messages and Submissions' area above